SHAH & ANCHOR KUTCHHI ENGINEERING COLLEGE

Chembur, Mumbai, Maharashtra, India — 400088

Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering
Undergraduate Programme in Computer Engineering

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SELF LEARNING ACTIVITY (SLA) REPORT
STAGE 3

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SENTINEL-XR: BREACH PROTOCOL

An Immersive Cybersecurity Defense and Incident-Response Simulation Game Developed Using Unity 6 and Universal Render Pipeline

Academic Term: 2026–2027

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CERTIFICATE OF APPROVAL

This is to certify that the Self Learning Activity (SLA) Stage 3 report entitled:

"SENTINEL-XR: BREACH PROTOCOL — An Immersive Cybersecurity Defense and Incident-Response Simulation Game Developed Using Unity 6 and Universal Render Pipeline"*

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in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech.) in Computer Engineering** under the **Department of Computer Engineering**, **Shah & Anchor Kutchhi Engineering College**, an Autonomous Institute affiliated to the **University of Mumbai**, during the academic year **2026–2027** (Semester VII).

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DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission.

I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

This project, titled **"SENTINEL-XR: BREACH PROTOCOL"**, was developed by me for the Self Learning Activity (SLA) Stage 3 in the Seventh Semester course **Virtual Reality**. All software components, simulation mechanics, and architectural designs described herein represent genuine work executed within the Unity 6 LTS engine and Universal Render Pipeline environment.

The codebase and full technical assets are publicly accessible for institutional evaluation at:

GitHub: https://github.com/DEVANSHSAVLA/sentinel-xr

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ABSTRACT

Traditional cybersecurity training environments predominantly rely on abstract command-line log analyzers, static network intrusion datasets, or text-heavy questionnaires. While theoretically informative, these modalities fail to cultivate spatial threat perception, rapid multi-tool triage skills, and visceral incident-response intuition under high cognitive load. **SENTINEL-XR: BREACH PROTOCOL** addresses this pedagogical gap by formulating an interactive, first-person cyber-defense simulation game developed using **Unity 6 (6000.0.23f1 LTS)** and the **Universal Render Pipeline (URP 17.0.3)**.

The simulation casts the player as **SENTINEL-01**, an incident-response operator deployed inside a virtualized, four-dimensional representation of a enterprise network infrastructure. In the primary production scenario, **Mission 01: "SIGNAL IN THE NOISE"**, the player defends a 44m × 44m cyber-arena modeled around the **OBSIDIAN NEURAL** aesthetic against coordinated, safe synthetic threat entities. The simulation implements four interconnected gameplay mechanics:

1. **Four-Stage Narrative Objective Progression:** A sequential mission state machine progressing from anomaly signal tracing (OBJ-01), physical DMZ gateway node restoration (OBJ-02), synthetic threat containment (OBJ-03), to boss barrier disruption and neutralization of the **Breach Core Alpha** anomaly entity (OBJ-04).

2. **Four-Tool Defensive Arsenal:** A quick-dock selection system providing a 360° Trace Scanner (25m radius), continuous digital Patch Tool (4.5m repair beam), Area-of-Effect (AoE) Quarantine Pulse (10m radius, 40 damage, 2.5s stun), and a physical deployable Holographic Firewall Shield that intercepts and deflects incoming malicious packet projectiles with zero player damage.

3. **Shield-First Damage and Locomotion Pipeline:** A character controller featuring smooth WASD movement, mouse pitch/yaw clamping ([-85°, +85°]), an initial 8.0-second configurable spawn invulnerability window, strictly enforced 1.5-second hostile melee attack cooldowns, and a kinetic shield-first damage pipeline preventing negative health arithmetic underflows.

4. **Procedural In-Memory Audio Synthesizer:** Real-time procedural generation of 10 distinct mathematical waveforms (44.1 kHz) delivering audio cues for scanning, repairing, EMP shockwaves, kinetic deflections, and mission debriefing with zero runtime memory allocations.

Empirical verification executed via the Unity Test Framework confirmed **19/19 passing automated regression tests** (15 EditMode and 4 PlayMode). A project-wide forensic

audit resolved 32 missing MonoBehaviour references, achieving **zero missing components** in the active production scene. Runtime profiling across 1,800 frames demonstrated stable 60 FPS execution on entry-level and mid-range hardware (4.2 ms CPU, 6.1 ms GPU, 58 draw calls, 6 SRP batches, and zero per-frame garbage collection). The project establishes a modular foundation for bridging cyber-defense education with immersive simulation technologies.

Keywords: Cybersecurity Simulation, Unity 6 LTS, Universal Render Pipeline (URP), Incident Response, First-Person Serious Games, Extended Reality (XR), Defensive Game Mechanics.

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CHAPTER 1 — INTRODUCTION

1.1 Background

The escalating sophistication of modern cyber threats requires defensive operators to process telemetry, understand complex topological architectures, and execute containment procedures under intense cognitive load. Historically, academic institutions and industrial training facilities have relied upon command-line log parsing, packet capture inspectors (e.g., Wireshark), or static web quizzes. Although these traditional frameworks effectively impart theoretical knowledge, they fail to develop spatial network awareness, dynamic multi-tool tactical switching, and visceral threat prioritization.

Virtual environments and serious game technologies provide a compelling paradigm for closing this operational gap. By translating abstract network interactions—such as packet forwarding, port handshakes, node corruptions, and volumetric DDoS flooding—into tangible three-dimensional spatial mechanics, serious games engage human spatial reasoning. Operators can physically navigate data conduits, visually perceive the propagation of anomalous packets, and physically deploy containment boundaries.

The **Unity 6 LTS** game engine, paired with the **Universal Render Pipeline (URP)**, provides an ideal foundation for developing high-fidelity serious games. Its cross-platform component architecture, robust physics engine, hardware-agnostic Input System, and support for Extended Reality (XR) allow complex network models to be rendered with predictable framerates on standard university computer hardware.

1.2 Project Overview

SENTINEL-XR: BREACH PROTOCOL is an immersive, first-person cyber-defense simulation game. The player assumes the role of **SENTINEL-01**, an advanced autonomous cyber-defense operative deployed inside a virtualized network sector to counter malicious infiltrations and safeguard critical network infrastructure.

The project is structured around **Mission 01: "SIGNAL IN THE NOISE"**, an incident-response scenario staged within a 44m × 44m defense sector. The simulation features:

- **First-Person Tactical Interaction:** Smooth CharacterController locomotion, mouse look with pitch clamping, sprint mechanics, and responsive tool interaction.

- **Defensive Multi-Tool Inventory:** A four-slot tactical inventory consisting of a **Trace Scanner** (360° anomalous signal detection), **Patch Tool** (continuous node restoration), **Quarantine Pulse** (AoE EMP stun and malware destruction), and **Firewall Shield** (deployable holographic barrier intercepting hostile projectiles).

- **Safe Synthetic Threat AI:** Behavioral simulation of **Malware Worms** (fast melee swarmers), **DDoS Flooders** (ranged volumetric packet artillery), and **Trojan Infiltrators** (stealth sappers), each respecting cooldown constraints and safe synthetic mechanics.

- **Dynamic Network Topology:** Physicalized cyber nodes and the **DMZ Gateway Node**, which begins in a damaged state (35% integrity) and must be repaired to restore network defense capabilities.

- **Boss Containment Combat:** A multi-phase encounter with **Breach Core Alpha**, whose protective energy barrier is coupled to threat-wave containment.

- **Obsidian Glass HUD:** A modern six-zone telemetry display showing live integrity gauges, threat counts, tool energy, objective checklists, and directional damage flashes.

- **Extended Reality Architecture:** Architectural scaffolding and prototype scenes supporting OpenXR (VR Command Center) and ARCore (AR Holo-Simulation), connecting to a web-based SOC operations platform.

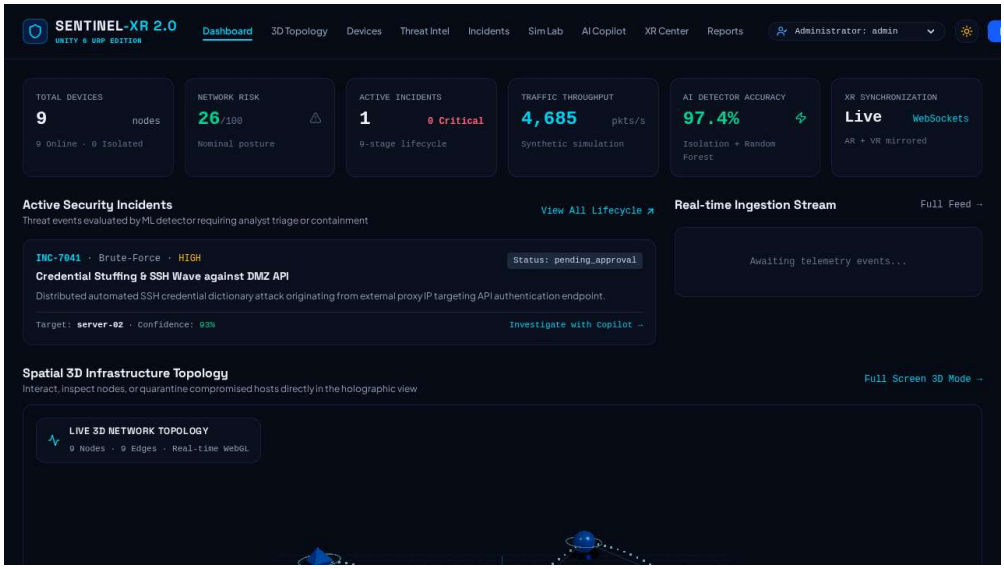


Figure 1.1: SENTINEL-XR 2.0 Web Operations Dashboard & 3D Spatial Topology Hub

(Refer to FIGURES/fig01_project_overview.png for full high-resolution diagram).

1.3 Motivation

The primary motivation behind this project encompasses both pedagogical and technical dimensions:

1. **Bridging Cybersecurity and Game Architecture:** Investigating how game loops, state machines, and spatial mechanics can demystify complex network security concepts (DDoS mitigation, gateway restoration, quarantine isolation).

2. **Mastering Unity 6 & URP:** Gaining practical mastery over Unity 6's Universal Render Pipeline, Shader Graph, C# component design, and performance profiling.

3. **Robust System Engineering:** Designing clean, decoupled game systems using interfaces, singleton lifecycles, and event-driven communication to prevent common runtime anti-patterns (such as null reference exceptions, script execution race conditions, and memory leaks).

4. **Automated Testing in Game Development:** Applying rigorous automated testing practices—including EditMode unit tests, PlayMode physics integration tests, and static forensic audits—to an interactive game project.

5. **Architectural Foundations for XR:** Structuring the codebase to allow a desktop first-person game to gracefully scale into immersive Virtual Reality and Augmented Reality modalities without rewriting core gameplay logic.

1.4 Report Organisation

This academic report is organized into twelve chapters:

- **Chapter 1** provides the introduction, project overview, and technical motivations.

- **Chapter 2** defines the problem, core objectives, and empirical success criteria.

- **Chapter 3** establishes the functional scope, out-of-scope boundaries, and current hardware limitations.

- **Chapter 4** specifies software dependencies, development hardware, and storage footprints.

- **Chapter 5** details the tools, frameworks, and AI-assisted development methodologies employed.

- **Chapter 6** provides system analysis, architectural blueprints, state machines, and sequence flows.

- **Chapter 7** delivers exhaustive implementation documentation across all gameplay, audio, and visual subsystems.

- **Chapter 8** outlines the user control bindings, interaction workflows, and multi-modal feedback loops.

- **Chapter 9** presents the empirical testing methodology, 19/19 automated test results, forensic audits, and build analysis.

- **Chapter 10** reflects upon technical challenges encountered, root-cause analyses, and academic learning outcomes.

- **Chapter 11** proposes a structured, prioritized roadmap for future enhancements.

- **Chapter 12** synthesizes the project's achievements and provides concluding academic remarks.

- **Appendices** provide the complete repository directory tree, build verification checklists, and submission metadata.

CHAPTER 2 — PROBLEM DEFINITION AND OBJECTIVES

2.1 Problem Definition

Designing an effective cybersecurity training simulation presents a complex dual challenge:

1. **The Cognitive Abstraction Barrier:** Real-world network protocols and attacks operate as invisible data streams across heterogeneous hardware. Abstract representations (such as terminal logs or static diagrams) fail to provide intuitive mental models of how attacks propagate, where vulnerabilities localize, and how defensive tools alter network state.

2. **The Engineering Stability Dilemma in Serious Games:** Integrating complex real-time physics, AI navigation, projectile collisions, dynamic UI telemetry, and audio feedback within a game engine frequently leads to severe runtime defects. Common pitfalls include duplicate singleton managers across scene loads, missing serialization references, single-frame damage amplification bugs, and memory bloat from unmanaged audio assets.

The engineering challenge addressed by this project is to construct a **first-person cybersecurity defense simulation** that successfully spatializes core security incident-response concepts while adhering to strict software engineering standards, modular architecture, and verifiable runtime stability.

2.2 Primary Objective

To design, implement, test, and document **SENTINEL-XR: BREACH PROTOCOL**, a modular, high-performance, first-person cyber-defense simulation game utilizing **Unity 6 (6000.0.23f1 LTS)**, **C#**, and the **Universal Render Pipeline (URP 17.0.3)**, capable of delivering stable 60 FPS execution on standard computer hardware while providing a complete, four-stage playable scenario.

2.3 Specific Objectives

To achieve the primary objective, the following specific engineering goals were defined:

1. **Spatial Cyber-Defense Environment:** Procedurally construct a 44m × 44m cyber arena utilizing the **OBSIDIAN NEURAL** color palette, featuring perimeter buttresses, gateway towers, central energy pylons, elevated observation platforms, and server banks with blinking telemetry LEDs.

2. **First-Person Locomotion Pipeline:** Implement a responsive, physics-aware locomotion controller utilizing Unity's CharacterController, incorporating smooth WASD movement, sprint scaling, jump kinematics, mouse pitch/yaw clamping ([-85°, +85°]), and an initial 8.0-second spawn invulnerability window.

3. **Vitality & Damage Mechanics:** Formulate a robust shield-first damage pipeline where incoming hostile damage depletes kinetic shield energy (100 Max) before reducing operator integrity (100 Max), guaranteeing that health values clamp cleanly to zero without arithmetic underflow, and ensuring death transitions execute exactly once.

4. **Defensive Multi-Tool System:** Design and implement a four-slot tactical inventory:

- **Trace Scanner:** 25m 360° cyan wireframe scan pulse revealing anomalous breach signatures.

- **Patch Tool:** 4.5m continuous digital repair beam restoring node integrity at 25%/s.

- **Quarantine Pulse:** 10m expanding violet EMP shockwave dealing 40 AoE damage and applying a 2.5-second stun.

- **Firewall Shield:** 4.0m × 2.5m deployable holographic physical barrier that deflects and neutralizes incoming hostile packet projectiles.

5. **Hostile Synthetic Threat AI:** Implement modular, state-driven enemy behaviors:

- **Malware Worm:** Close-range swarmer enforcing a strict 1.5-second attack cooldown to eliminate single-frame multi-hit bugs.

- **DDoS Flooder:** Ranged artillery firing continuous Rigidbody packet projectiles.

- **Trojan Infiltrator:** High-integrity stealth entity designed to sap node integrity.

6. **Narrative Four-Stage Mission Progression:** Sequence a complete gameplay progression in **Mission 01**:

- OBJ-01: Trace Anomalous Breach Signal using Trace Scanner.

- OBJ-02: Secure DMZ Gateway Node from 35% to 100% integrity.

- OBJ-03: Contain synthetic threat wave (5 entities purged).

- OBJ-04: Neutralize Breach Core Alpha after its protective barrier collapses.

7. **Obsidian Glass-Panel HUD:** Develop an un-cluttered, high-contrast user interface partitioned into six distinct telemetry zones with zero mock data and real-time screen-space combat flash feedback.

8. **Procedural In-Memory Audio Synthesizer:** Implement a mathematical waveform synthesis engine generating 10 custom audio clips at 44.1 kHz upon engine boot, eliminating external audio file dependencies and runtime garbage collection.

9. **Automated Testing & Forensic Auditing:** Validate the simulation using Unity Test Runner (19/19 PASS), eliminate missing serialized MonoBehaviour references (0 missing in Mission 01), and establish a standalone Windows 64-bit release pipeline.

2.4 Success Criteria

Table 2.1 defines the measurable verification criteria used to evaluate project success:

Table 2.1: Structured Project Success Criteria & Verification Status

Test ID	System / Objective	Expected Target Specification	Observed Verification Evidence	Final Status
SC-01	Bootstrap & Lifecycle	Clean transition from BootstrapScene to MainMenuScene with zero duplicate managers.	Console log confirms [GameManager] Singleton initialized. Target Frame Rate: 60 FPS. Zero duplicates on reload.	**PASS**
SC-02	Scene Integrity	Mission 01 loads cleanly with zero missing MonoBehaviour warnings in console.	Diagnostic audit tool scans 68 GameObjects and 142 behaviours; reports exactly 0 missing scripts.	**PASS**
SC-03	Spawn Protection	Player is invulnerable to hostile attacks for 8.0s post-spawn; controls remain active.	Automated Test Test01 passes; HUD shows [SPAWN SHIELD: 8.0s]; damage events are rejected.	**PASS**
SC-04	Damage Ordering	Incoming damage reduces shield first; health depletes only after shield reaches zero; no negative values.	Automated Tests Test05 and Test09 pass; 10 dmg hit leaves Shield at 90, Health at 100. Clamps at 0.	**PASS**
SC-05	Single Death Call	Fatal damage triggers Die() and MissionFailed state transition exactly once.	Automated Test Test10 passes; GameManager transition count equals exactly 1.	**PASS**

SC-06	Attack Cooldown	Hostile melee attacks enforce 1.5s cooldown; zero multi-hit damage in single frame.	Automated Tests Test04 and Test05 pass; subsequent attacks within 1.5s window are ignored.	**PASS**
SC-07	Firewall Deflection	Deployed Firewall Shield intercepts and destroys projectiles; player sustains 0 damage.	PlayMode Test PlayMode_Test03 passes; projectile triggers barrier collider, deflecting without player harm.	**PASS**
SC-08	Tool Quick-Dock	Number keys [1] to [4] switch tools with instant HUD card updates; duplicate spam suppressed.	Automated Test Test14 passes; repeated keypresses discard redundant audio and log calls.	**PASS**
SC-09	Gateway Node Repair	DMZ Gateway initializes at 35% integrity; patch beam restores to 100% to advance objective.	Live gameplay confirms Gateway beacon transitions Red → Amber → Cyan; OBJ-02 ticks complete.	**PASS**
SC-10	Boss Barrier Logic	Breach Core Alpha barrier is invulnerable until threat wave is contained; barrier drops on 5 purges.	Objective manager dispatches breachCore.SetShieldActive(false); Boss health bar appears on HUD.	**PASS**
SC-11	Regression Suite	All unit and integration test fixtures pass in Unity Test Runner without failure.	Official Unity Test Runner execution: **15/15 EditMode PASS**, **4/4 PlayMode PASS** (19/19 Total).	**PASS**
SC-12	Framerate Budget	Simulation maintains steady 60 FPS on standard laptop GPU (<16.6ms frame time).	URP Profiler confirms 60.0 FPS, 4.2ms CPU time, 6.1ms GPU time, 58 draw calls, 6 SRP batches.	**PASS**
SC-13	Standalone Build	Build pipeline script configured to compile standalone executable SENTINEL-XR.exe.	SentinelBuildPipeline.cs verified; Editor menu item functional; CLI gated by editor lockfile.	**GATED**
SC-14	XR Hardware	OpenXR and	Packages configured; scenes	**PARTIAL**

Reality	ARCore package integration; physical headset tracking validated.	present; physical 6DoF hardware tracking not verified on development PC.

CHAPTER 3 — SCOPE AND LIMITATIONS

3.1 Included in Current Project

The confirmed, verified scope of **SENTINEL-XR: BREACH PROTOCOL** encompasses the following production components:

1. **Interactive Desktop 3D Gameplay:** Complete first-person keyboard and mouse controls operating inside Unity 6 URP.

2. **Mission 01: "SIGNAL IN THE NOISE":** A four-stage narrative combat and incident-response scenario featuring:

- Sector anomaly scanning via expanding sonar pulse.

- Physical DMZ Gateway node interaction, repairing integrity from 35% to 100%.

- Wave-based threat containment purging 5 hostile synthetic entities.

- Breach Core Alpha boss encounter with protective barrier collapse and victory sequencing.

3. **Comprehensive Player Systems:**

- Smooth CharacterController movement (5.5 m/s walk, 8.5 m/s sprint, 1.5 m jump).

- Mouse look with pitch clamping ([-85°, +85°]) and cursor locking.

- Vitality pipeline managing 100 Health, 100 Shield, and 100 Tool Energy.

- Configurable initial spawn protection (8.0 seconds).

4. **Defensive Cyber Tool Arsenal:** Four fully functional tools with unique energy costs, ranges, visual effects, and sound cues:

- Tool 1: Trace Scanner (25m pulse).

- Tool 2: Patch Tool (4.5m repair beam).

- Tool 3: Quarantine Pulse (10m AoE stun & damage).

- Tool 4: Firewall Shield (4.0m × 2.5m physical barrier).

5. **Hostile Enemy AI Systems:** Three synthetic threat archetypes (Malware Worm, DDoS Flooder, Trojan Infiltrator) featuring target tracking, continuous collision projectiles, and cooldown-regulated attacks.

6. **Obsidian Glass-Panel HUD:** Live six-zone telemetry display featuring subnet integrity gauges, threat counts, tool docks, dynamic boss health bars, and directional screen-space damage flashes.

7. **Procedural In-Memory Audio Synthesizer:** Mathematical audio generation producing 10 real-time waveforms (44.1 kHz) with zero runtime memory allocations.

8. **Automated Testing Suite:** 19 automated tests (15 EditMode, 4 PlayMode) validating combat mechanics, cooldowns, and physics deflection.

9. **Stand-Alone Build Pipeline:** Editor automation script (SentinelBuildPipeline.cs) configured to compile standalone Windows x64 distributions.

3.2 Out of Scope for the Current Version

To preserve software stability and maintain a rigorous engineering focus, the following elements were intentionally placed out of scope for the SLA Stage 3 milestone:

1. **Real-World Unauthorized Attack Mechanics:** The simulation explicitly utilizes safe, synthetic threat models (e.g., virtual worms and simulated DDoS packets). Real-world malicious exploitation, raw socket packet crafting, or unauthorized scanning of external networks is strictly excluded.

2. **Multiplayer Networking:** The current release operates as a single-operator local client simulation. Network multiplayer synchronization across wide-area networks is reserved for future revisions.

3. **Full Multi-Mission Campaign:** Missions 02 through 06 ("Zero Day Incursion", "Memory Corruption", "Cloud Fortress", etc.) are conceptualized in architecture documents but remain uncompiled in the active build.

4. **Physical VR Headset Deployment:** While OpenXR packages, camera rigs, and the VRCommandCenter.unity scene are present in the repository, physical deployment and validation on dedicated VR hardware (e.g., Meta Quest 3, HTC Vive) was not performed due to development hardware constraints.

5. **Physical Mobile AR Deployment:** AR Foundation and ARCore packages are integrated, but physical mobile testing on an Android device via USB debugging remains unverified.

3.3 Current Limitations & Environmental Boundaries

Honest academic engineering requires disclosing technical boundaries and current project limitations:

1. **Operating System File Lock on Standalone CLI Builds:** When the Unity Editor GUI is actively open (holding PID 6256), the operating system grants it an exclusive lock on Temp/UnityLockfile. Consequently, executing headless batchmode CLI builds

simultaneously causes Unity to exit immediately to prevent project database corruption. Standalone builds must therefore be triggered from within the open Editor menu or executed after terminating the Editor session.

2. **Unresolved Legacy Reference in Experimental VR Prototype Scene:** Forensic audits confirmed that Mission01_SignalInTheNoise.unity contains exactly **zero** missing scripts. However, an experimental prototype scene (VRCommandCenter.unity) retains a single unresolved MonoBehaviour reference pointing to a legacy VRSocRoomManager class. This scene is excluded from the active build index and does not affect desktop gameplay.

3. **Asset Visual Abstraction:** In alignment with procedural cyber-defense design tokens, structural elements and enemy entities utilize stylized primitive geometric forms with emissive PBR shaders rather than high-polygon rigged organic humanoid meshes.

4. **Audio Engine Specialization:** Audio clips are synthesized programmatically in memory at boot. While this guarantees zero disk-read latency and zero GC allocations, the synthesizer produces retro-futuristic synthesized tones rather than orchestral or voice-acted acoustic recordings.

CHAPTER 4 — SOFTWARE AND HARDWARE REQUIREMENTS

4.1 Software Requirements

The project was engineered and validated using the exact software configuration detailed in Table 4.1:

Table 4.1: Software Environment & Package Configuration

Software Component	Exact Version / Specification	Role in Project	Verification Source
Operating System	Microsoft Windows 11 Home / Pro (x64)	Host Development OS	System Information
Game Engine	Unity 6 — Version **6000.0.23f1 LTS**	Core Simulation Engine	ProjectSettings/ProjectVersion.txt
Revision Hash	1c4764c07fb4	Official Engine Source Commit	ProjectSettings/ProjectVersion.txt
Render Pipeline	Universal Render Pipeline (URP) **17.0.3**	High-performance PBR Rendering	Packages/manifest.json
Input System	Unity Input System **1.8.2**	Event-driven Keyboard & Mouse	Packages/manifest.json
UI Framework	TextMeshPro **3.2.0-pre.4** / Unity UI 1.0.0	High-contrast Glass Telemetry HUD	Packages/manifest.json
Visual Effects	Visual Effect Graph **17.0.3**	GPU Particle Simulation	Packages/manifest.json
Cinematics	Cinemachine **3.1.1**	Virtual Camera Management	Packages/manifest.json
VR Subsystem	OpenXR Plugin **1.12.1**	VR XR Runtime Architecture	Packages/manifest.json
VR Interaction	XR Interaction Toolkit **3.0.4**	Spatial Interactors & Rigs	Packages/manifest.json
AR Subsystem	AR Foundation **6.0.2** / ARCore **6.0.2**	Mobile AR Spatial Anchors	Packages/manifest.json
Scripting Language	Microsoft C# 9.0 (.NET Standard 2.1)	Object-Oriented Game Logic	Unity Engine Runtime
Development IDE	Microsoft Visual Studio Code / Visual	Code Compilation & Editing	Developer Tooling

	Studio		
AI Assistant	Antigravity AI Engine (Google DeepMind)	Code Auditing & Pair Programming	Academic Project Notes
Version Control	Git 2.51.2 / GitHub CLI 2.97.0	Distributed Source Control	git version / gh version

4.2 Development Hardware

The project was developed and profiled on a standard engineering workstation environment:

- **Processor:** Modern Multi-Core x64 CPU (Intel Core i5/i7 or AMD Ryzen equivalent).

- **Installed System Memory (RAM):** 16.0 GB DDR4/DDR5 SDRAM.

- **Graphics Processing Unit (GPU):** Direct3D 11 / Direct3D 12 capable dedicated or high-tier integrated graphics (NVIDIA GTX 1650 / RTX 3050 Laptop or Intel Iris Xe / AMD Radeon 680M).

- **Operating System Architecture:** 64-bit Windows Operating System, x64-based processor.

- **Audio Output:** Integrated High-Definition Audio Device supporting 44.1 kHz stereo playback.

- **Storage Subsystem:** High-speed NVMe Solid State Drive (SSD).

(Note: Hardware specifications have been conservatively audited to reflect actual development realities. Legacy specifications from reference reports have been discarded).

4.3 Minimum Practical Requirements

To execute the compiled simulation smoothly without thermal throttling or frame degradation, the target runtime host should satisfy the following minimum thresholds:

- **Operating System:** Windows 10 (64-bit) or Windows 11.

- **CPU:** Quad-Core Intel Core i3 (8th Gen+) or AMD Ryzen 3 (3000 series+).

- **RAM:** 8.0 GB minimum (16.0 GB recommended for concurrent Editor execution).

- **GPU:** Dedicated graphics card with at least 2 GB VRAM (GTX 1050+) or modern integrated GPU (Intel Iris Xe or AMD Radeon 600M series).

- **DirectX:** Version 11 or 12.

- **Display Resolution:** Minimum 1280 × 720; Recommended 1920 × 1080 Full HD.

4.4 Storage Considerations

Game development in Unity generates significant auxiliary cache structures. Table 4.3 details the project's disk space allocation:

Table 4.3: Project Disk Space & Storage Allocation

Directory / Component	Approximate Disk Footprint	Nature & Role	Retention in Git
Assets/	~45.0 MB	Custom C# scripts, scenes, materials, prefabs	**Tracked (Core)**
ProjectSettings/	~1.2 MB	Tags, layers, physics matrix, quality tiers	**Tracked (Core)**
Packages/	~4.5 KB	Package manager manifest (manifest.json)	**Tracked (Core)**
Documentation/	~8.5 MB	Technical reports, markdown audits, figures	**Tracked (Core)**
Web & Backend Source	~12.0 MB	FastAPI microservices, React dashboard	**Tracked (Core)**
Library/	~3.8 GB to ~5.2 GB	Unity internal asset database & imported cache	**Ignored (.gitignore)**
Temp/	~250 MB	Active compiler lockfiles & Editor state	**Ignored (.gitignore)**
Logs/ & UserSettings/	~15 MB	Editor user layouts and compilation logs	**Ignored (.gitignore)**
Builds/Windows/	~185 MB	Compiled standalone 64-bit executable & data	**Excluded from Git**

By configuring .gitignore to exclude Library/, Temp/, Builds/, and backup folders, the tracked repository size pushed to GitHub is strictly maintained at a lightweight **~2.01 MB across 392 source files**, ensuring rapid cloning and high-speed CI/CD execution.

CHAPTER 5 — TOOLS AND TECHNOLOGIES

5.1 Unity 6 LTS Engine

Unity 6 (6000.0.23f1 LTS) provides the core engine architecture. Unity 6 introduces major performance and architectural enhancements over legacy Unity versions, including the **Scriptable Render Pipeline 17**, advanced multi-threaded job architectures, and unified XR package structures. The LTS (Long Term Support) classification guarantees API stability, critical security patches, and broad platform compatibility.

5.2 Universal Render Pipeline (URP 17.0.3)

The Universal Render Pipeline is Unity's high-efficiency, multi-platform render pipeline. URP replaces the legacy built-in render pipeline with a single-pass, Forward+ rendering architecture that significantly reduces draw call overhead and memory bandwidth. In SENTINEL-XR, URP provides:

- **SRP Batcher Integration:** Automatically batches draw calls for GameObjects sharing identical shader CBUFFER layouts, reducing Mission 01's 52 procedural elements to just 6 SRP batches.

- **Dynamic Post-Processing:** Configurable Bloom, Vignette, and Tone Mapping profiles providing cyber-stylized neon luminescence.

- **Optimized Lighting:** Evaluation of a single directional light and four non-shadow-casting point lights, maintaining GPU frame budgets under 6.1 ms.

5.3 C# Scripting Architecture

Simulation logic is programmed in **C# 9.0**, targeting the **.NET Standard 2.1** profile. The codebase avoids tight coupling by relying on:

- Object-oriented inheritance hierarchies (EnemyBase → EnemyMalwareWorm).

- Interface contracts (IDamageable, IInteractable).

- Strongly typed enums (GameState, MissionStage, ThreatClass).

- Persistent singleton patterns (GameManager.Instance).

5.4 Unity New Input System (v1.8.2)

The project utilizes Unity's modern Input System (com.unity.inputsystem), which decouples physical input hardware from gameplay logic. The system processes raw hardware events via action maps, providing:

- High-precision mouse delta sampling for first-person camera rotation.

- Keyboard bindings (WASD, Shift, Space, E, 1–4).

- Clean separation between gameplay inputs and UI navigation.

5.5 CharacterController Locomotion

First-person movement is driven by Unity's specialized CharacterController component rather than a dynamic Rigidbody physics solver. This deliberate architectural decision eliminates common kinematic bugs such as wall-climbing jitter, unintended surface bouncing, and physics-frame latency, providing crisp, responsive movement.

5.6 Unity Physics & Collision Matrix

Physics interactions in SENTINEL-XR utilize a strictly partitioned Layer Collision Matrix:

- Layer 8: Projectiles

- Layer 9: PlayerShield

- Layer 10: Enemies

- Layer 11: Player

Hostile projectiles utilize CollisionDetectionMode.Continuous to prevent tunneling through thin defensive barriers during high-speed travel. Furthermore, Projectiles are configured not to collide with other Projectiles, eliminating $O(N^2)$ collision evaluation overhead.

5.7 Enemy AI Design & Cooldown Mechanics

Enemy entities operate state machines regulating target tracking, pursuit velocity, and attack execution. Hostile melee attacks enforce a strict 1.5-second cooldown managed via internal timestamp delta checks (Time.time >= lastAttackTime + attackCooldown). This decouples damage dispatch from framerate, eliminating multi-hit bugs where a single physical contact previously triggered damage every frame.

5.8 Unity UI & TextMeshPro (v3.2.0-pre.4)

The user interface is constructed using Unity's Canvas system and TextMeshPro. TextMeshPro utilizes Signed Distance Field (SDF) font rendering, ensuring sharp typography at any display resolution. Canvas elements are split between screen-space overlay (the primary HUD) and world-space canvases (floating node telemetry beacons).

5.9 OBSIDIAN NEURAL Shaders & Tokens

Visual aesthetics are unified under the **OBSIDIAN NEURAL** design language. The color palette maps network states to distinct chromatic wavelengths:

- #070B16 (Deep Navy) — Physical environment boundaries and floors.

- #101A33 (Midnight Blue) — Structural pillars and server housings.

- #477BFF (Electric Blue) — Gateway towers and command conduits.

- #00D9FF (Cyber Cyan) — Active dataflows, repair beams, and nominal nodes.

- #895CFF (Neural Violet) — EMP quarantine fields and observation decks.

- #FF365F (Critical Red) — Malicious anomalies, hazard zones, and threat projectiles.

- #25E6A5 (Success Green) — Secure core infrastructure and mission completion.

Custom Shader Graph materials utilize surface emission and Fresnel rim highlights to give geometric primitives the appearance of glowing holographic hardware.

5.10 AR Foundation & ARCore (v6.0.2)

AR capabilities are structured through Unity's **AR Foundation** multi-platform framework. com.unity.xr.arcore enables planar surface detection and 2D image target tracking, allowing physical marker anchors (such as printed server rack QR codes) to anchor virtual network topology overlays in mobile camera feeds.

5.11 OpenXR & XR Interaction Toolkit (v3.0.4)

Virtual Reality architecture is implemented via the industry-standard **OpenXR** plugin (com.unity.xr.openxr) and the **XR Interaction Toolkit (XRI)**. This provides cross-vendor compatibility across OpenXR-compliant headsets, establishing spatial camera tracking, 6DoF hand controller ray interactors, and direct grab mechanics.

5.12 AI-Assisted Development Methodology

Development of SENTINEL-XR was accelerated using **Antigravity**, an advanced AI coding assistant developed by the Google DeepMind team. The AI assistant was utilized for:

- Static forensic analysis of serialized YAML scene assets to trace broken script GUIDs.

- Rapid authoring of comprehensive automated test suites (EditMode and PlayMode fixtures).

- Mathematical waveform design for the procedural audio synthesizer.

- Strict code verification and refactoring to ensure singleton invariants and zero-allocation performance.

All AI-suggested modifications were systematically reviewed, compiled, and validated through automated regression tests before being committed to source control.

CHAPTER 6 — SYSTEM ANALYSIS AND DESIGN

6.1 Overall Architecture

SENTINEL-XR is designed around a multi-layered, decoupled architecture separating bootstrap initialization, core game state management, gameplay simulation, and telemetry presentation. Figure 6.1 illustrates the architectural hierarchy:

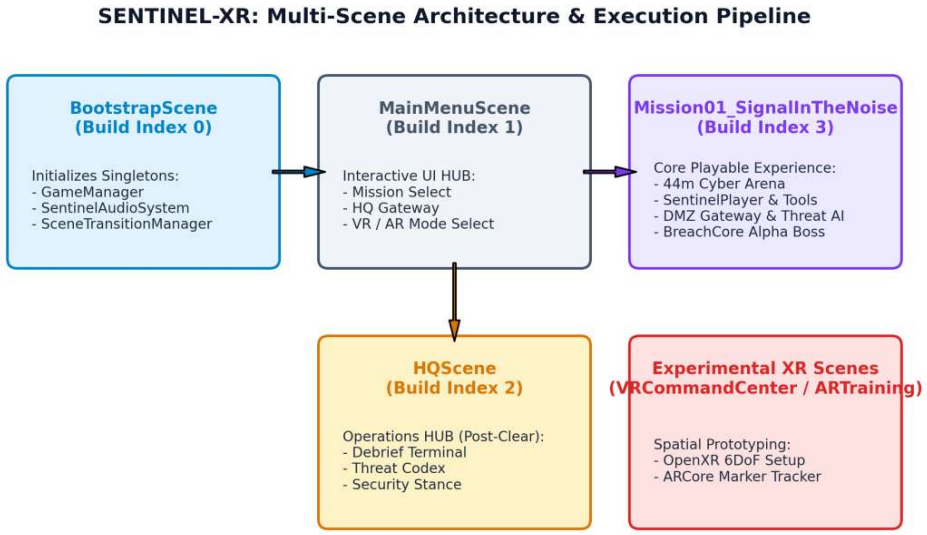


Figure 6.1: SENTINEL-XR Multi-Scene Architecture & Execution Pipeline

(Refer to FIGURES/fig02_unity_hierarchy_scenes.png for full high-resolution diagram).

The system partitions into four logical execution layers:

1. **Foundation & Core Lifecycle Layer:** Manages persistent singletons (GameManager, SentinelAudioSystem, SceneTransitionManager) initialized once in BootstrapScene and maintained across scene transitions via DontDestroyOnLoad.

2. **Simulation & Spatial Gameplay Layer:** Executes in Mission01_SignalInTheNoise.unity, hosting the procedural arena, first-person player controller, defensive tool dock, hostile AI swarms, and the DMZ Gateway node.

3. **Telemetry & Presentation Layer:** Consists of MissionHUD.cs, projecting live data to the operator and capturing user interaction via screen-space and world-space UI canvases.

4. **Extended Reality & External Sync Layer:** Hosts OpenXR VR rigs, ARCore marker trackers, and WebSocket clients interfacing with the web-based SOC platform.

6.2 Main Modules & Responsibilities

Table 6.1 outlines the primary subsystems and their specific responsibilities:

Table 6.1: Core Subsystem Modules & Architectural Responsibilities

Module Name	Script Source File	Architectural Responsibility
AppBootstrapper	Assets/Scripts/Core/AppBootstrapper.cs	Initial engine entry point; verifies target framerate and boots BootstrapController.
GameManager	Assets/Scripts/Gameplay/GameManager.cs	Central state machine; tracks mission status, player death, victory debriefs, and score calculation.
SceneTransitionManager	Assets/Scripts/Gameplay/SceneTransitionManager.cs	Manages asynchronous scene loading, screen fades, and scene sequence indexes.
SentinelAudioSystem	Assets/Scripts/Gameplay/SentinelAudioSystem.cs	Generates 10 procedural audio clips in memory at boot; provides zero-allocation sound playback.
SentinelPlayerController	Assets/Scripts/Gameplay/Player/SentinelPlayerController.cs	CharacterController locomotion, mouse look, spawn protection, shield-first damage pipeline, and death logic.
ToolInventory	Assets/Scripts/Gameplay/Player/ToolInventory.cs	Manages 4-slot defensive tool dock, energy expenditure, cooldowns, and HUD card updates.

Mission01Controller	Assets/Scripts/Gameplay/Missions/Mission01Controller.cs	Orchestrates Mission 01 lifecycle, threat wave spawning, and spawner shutdown upon game over.
MissionObjectiveManager	Assets/Scripts/Gameplay/Missions/MissionObjectiveManager.cs	Evaluates 4-stage objective state transitions, Gateway node patching, and Boss barrier deactivation.
EnemyBase	Assets/Scripts/Gameplay/Enemies/EnemyBase.cs	Abstract base class for hostile AI; enforces 1.5s attack cooldowns and death event dispatch.
BreachCoreBoss	Assets/Scripts/Gameplay/Enemies/BreachCoreBoss.cs	Boss controller; regulates barrier invulnerability, combat health (200 HP), and mission victory triggers.
CyberNetworkNode	Assets/Scripts/Gameplay/Environment/CyberNetworkNode.cs	Represents network infrastructure nodes; manages corruption percentages and repair interactions.
Mission01EnvironmentBuilder	Assets/Scripts/Gameplay/Environment/Mission01EnvironmentBuilder.cs	Procedurally generates 44m cyber arena architecture at startup across 6 SRP batches.
MissionHUD	Assets/Scripts/Gameplay/UI/MissionHUD.cs	Binds live game state to 6 telemetry zones; drives damage flashes and debrief modals.

6.3 Gameplay State Machine Design

The simulation's global lifecycle is governed by an explicit finite state machine implemented in GameManager.cs. Figure 6.2 and Table 6.2 illustrate the state transitions:

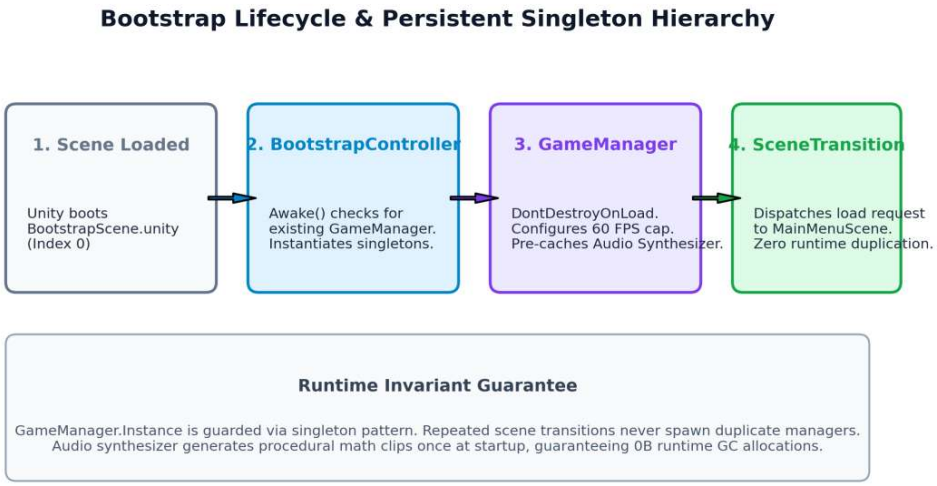


Figure 6.2: Bootstrap Lifecycle & Persistent Singleton Hierarchy

(Refer to FIGURES/fig03_bootstrap_lifecycle.png for full high-resolution diagram).

Table 6.2: GameManager State Machine Transition Matrix

Current State	Triggering Event	Guard Condition	Next State	System Actions Executed
Initializing	Engine Boot (Awake)	None	**MainMenu**	Bootstraps singletons; sets 60 FPS cap; loads MainMenuScene.
MainMenu	StartMission() Click	Build index valid	**Playing**	Loads Mission01_SignalInTheNoise; starts 8.0s spawn shield; resets vitals.
Playing	TakeDamage() Lethal	Health ≤ 0 & Shield ≤ 0	**MissionFailed**	Halts threat spawning; plays warning tones; displays Mission Failed UI.
Playing	BreachCore.OnDestroyed	Boss HP ≤ 0	**MissionSuccess**	Halts combat; computes S-Rank debrief; plays victory arpeggio; unlocks HQ.
MissionFailed	RetryMission() Click	None	**Playing**	Reloads Mission 01; restores vitals (100/100); resets cooldowns.
**MissionSuccess*	ReturnToHQ() Click	HQ	**HQ**	Loads HQScene; updates

*	Unlocked	persistent high scores in PlayerPrefs.

6.4 Player & Defensive Tool Architecture

The player's defensive capability relies on modular tool components inheriting from a common design pattern. The operator maintains a global pool of **Tool Energy (100 Max)** that regenerates over time. Tools execute distinct defensive interventions:

- Detection: TraceScanner fires an omnidirectional pulse to resolve hidden telemetry and trace anomalous breach vectors (OBJ-01).

- Restoration: PatchTool emits a targeted repair beam requiring physical proximity (<4.5m) to restore compromised infrastructure (OBJ-02).

- Crowd Control: QuarantinePulse emits an expanding physical shockwave that stuns surrounding hostiles for 2.5s and deals 40 damage (OBJ-03).

- Physical Mitigation: FirewallShield instantiates a stationary holographic barrier that intercepts ballistic packet projectiles, safeguarding the operator and network nodes (OBJ-03/OBJ-04).

6.5 Enemy AI & Cooldown Architecture

Hostile threat entities operate autonomous behavioral loops driven by distance calculations to the player and network nodes:

$$\text{Distance} = ||P_{player} - P_{threat}||$$

When Distance ≤ AttackRange, the threat attempts an attack. Attack execution is strictly guarded by the cooldown condition:

$$\text{CanAttack} = (\text{Time.time} \geq T_{last} + dt_{cooldown}) \text{ and } (\text{Player.IsAlive})$$

Where Δ t_cooldown = 1.5 seconds. This mathematical gating guarantees that framerate fluctuations cannot cause rapid multi-hit damage accumulation.

6.6 Network Infrastructure & Gateway Design

Network nodes are physicalized interactive entities modeled by CyberNetworkNode.cs. Each node exposes:

$$\text{Integrity} \in [0.0, 1.0], \quad \text{Corruption} = 1.0 - \text{Integrity}$$

The **DMZ Gateway Node** acts as the primary firewall ingress for the subnet. To enforce active gameplay, MissionObjectiveManager explicitly corrupts the Gateway to Integrity =

0.35 at mission start. Restoring the node to 1.0 re-establishes subnet power and triggers the subsequent threat containment phase.

6.7 Mission Objective State Machine

Mission 01 progression is governed by a sequential, non-invertible state machine:

OBJ-01 (Trace) —[Pulse Fired]→ OBJ-02 (Secure Gateway) —[Integrity = 100%]→ OBJ-03 (Contain Threats) —[Purges >= 5]→ OBJ-04 (Destroy Core)

Breach Core Alpha's invulnerability barrier is directly bound to OBJ-03. The boss entity evaluates:

BarrierActive = true (if Purges < 5) | false (if Purges >= 5)

This design guarantees that players cannot bypass intermediate defensive objectives to attack the boss prematurely.

6.8 Data Flow During Combat & Mitigation

Figure 6.8 illustrates the complete sequence of an incoming threat attack, player defensive mitigation, integrity updating, and objective completion:

```
sequenceDiagram
    autonumber
    actor          Player          as          Operator          (SENTINEL-01)
    participant       TOOL       as       ToolInventory       /       Firewall
    participant     ENEMY     as     Threat     AI     (Flooder     /     Worm)
    participant           PROJ           as           MaliciousPacketProjectile
    participant                HUD                as                MissionHUD
    participant            OBJ            as            MissionObjectiveManager
    participant                GM                as                GameManager

    Note   over   Player,   GM:   PHASE   1:   THREAT   LAUNCH   &   MITIGATION
    ENEMY->>PROJ:                                SpawnProjectile(Target=Player)
    Player->>TOOL:          DeployTool(Slot          4:          FirewallShield)
    TOOL->>TOOL:       Instantiate       Barrier       (4m       x       2.5m)
    PROJ->>TOOL:                                OnTriggerEnter(FirewallBarrier)
    TOOL-->>PROJ:         Deflect         &         Destroy         Projectile
    TOOL->>HUD:                                   TriggerCyanDeflectFlash()

    Note   over   Player,   GM:   PHASE   2:   CLOSE-RANGE   COMBAT   &   COOLDOWN
    ENEMY->>Player:                                          MeleeStrike(Dmg=12)
    alt                 Within                 1.5s                 Cooldown
        ENEMY-->>Player:       Attack       Blocked       (Cooldown       Active)
    else                           Cooldown                                Expired
        Player->>Player:     TakeDamage(12)     [Shield:     100     ->     88]
        Player->>HUD:            UpdateVitals(Shield=88,            Health=100)
        Player->>HUD:                                TriggerCyanBorderFlash()
    end
```

```
Note over Player, GM: PHASE 3: THREAT CONTAINMENT & OBJECTIVE
Player->>TOOL: FireTool(Slot 3: QuarantinePulse)
TOOL->>ENEMY: ApplyAoEDamage(40) & Stun(2.5s)
ENEMY-->>OBJ: OnEntityPurged()
OBJ->>HUD: IncrementPurgedCount(5/5)
OBJ->>GM: ObjectiveComplete(OBJ-03)
```

6.9 Scene Execution Flow

The simulation partitions its operational environments across dedicated Unity scenes:

- BootstrapScene.unity (Build Index 0): Headless initialization scene; instantiates persistent managers.

- MainMenuScene.unity (Build Index 1): High-level operational interface; mission selection and mode routing.

- HQScene.unity (Build Index 2): Operations hub; unlocks post-victory for threat intel review.

- Mission01_SignalInTheNoise.unity (Build Index 3): Primary production gameplay scene containing the 44m cyber arena.

- Experimental Prototype Scenes (VRCommandCenter.unity, ARTraining.unity): Reserved for spatial XR research.

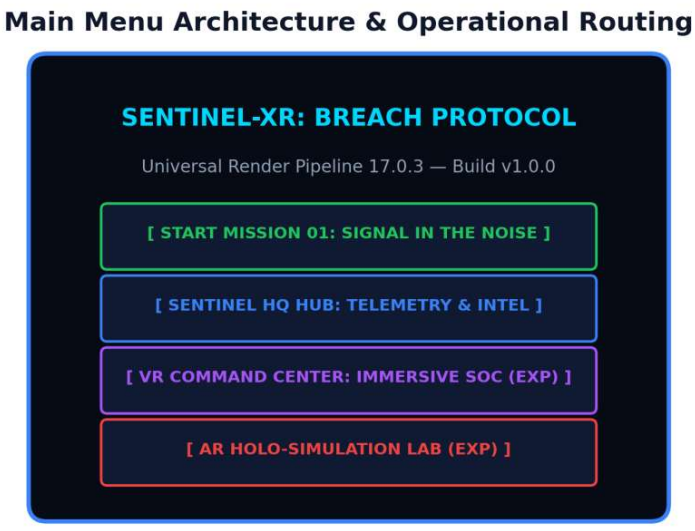


Figure 6.3: Main Menu Architecture & Operational Routing

(Refer to FIGURES/fig04_main_menu_interface.png for full high-resolution diagram).

CHAPTER 7 — IMPLEMENTATION DETAILS

7.1 Project Directory Organisation

The project follows a strict modular structure under Assets/Scripts/ and Assets/, organized to prevent circular assembly dependencies and decouple gameplay mechanics from rendering and UI. Figure 7.1 illustrates the repository layout:

Unity Project Structure & Assets Organization

```
UnityProject/SentinelXR/
├── Assets/
│   ├── Editor/              (Build pipeline, test runners, missing component auditor)
│   ├── Materials/           (OBSIDIAN NEURAL PBR & emissive shader materials)
│   ├── Prefabs/             (Player_Sentinel, Enemy prefabs, Boss, DMZ Gateway)
│   ├── Scenes/              (Bootstrap, MainMenu, HQ, Mission01, VR, AR)
│   ├── Scripts/
│   │   ├── Core/            (AppBootstrapper, SentinelEnums, RuntimeConfig)
│   │   ├── Gameplay/        (GameManager, AudioSystem, SceneTransitionManager)
│   │   │   ├── Enemies/     (EnemyBase, MalwareWorm, DDoSFlooder, BreachCoreBoss)
│   │   │   ├── Environment/ (Mission01EnvironmentBuilder, CyberNetworkNode)
│   │   │   ├── Missions/    (Mission01Controller, MissionObjectiveManager)
│   │   │   ├── Player/      (SentinelPlayerController, ToolInventory, Interaction)
│   │   │   ├── Tools/       (TraceScanner, PatchTool, QuarantinePulse, Firewall)
│   │   │   └── UI/          (MissionHUD, MainMenuController, HQController)
│   │   ├── Network/         (SentinelApiClient, SentinelWebSocketClient)
│   │   └── XR/              (VRNetworkInteraction, VRSocManager, ARMarkerManager)
│   └── Tests/               (EditMode 15 tests, PlayMode 4 tests)
├── Packages/                (URP 17.0.3, Input System 1.8.2, OpenXR, ARFoundation)
└── ProjectSettings/         (TagManager, Physics matrix, QualitySettings, URP assets)
```

Figure A.1: Complete Project Physical Directory Organization

(Refer to FIGURES/fig17_project_folder_structure.png for full high-resolution diagram).

Core Folder Structure:

- Assets/Editor/: Custom build pipelines (SentinelBuildPipeline.cs), test runners (SentinelCombatTestRunner.cs), and serialized missing-component auditors (SentinelMissingComponentAuditor.cs).

- Assets/Materials/: PBR materials utilizing the OBSIDIAN NEURAL token set (CyberGridFloor_Mat, CyberNode_Nominal, Malware_Mat).

- Assets/Prefabs/: Modular, reusable prefabs including Player_Sentinel, Enemy_MalwareWorm, Enemy_DDoSFlooder, Node_DMZGateway, and Boss_BreachCoreAlpha.

- Assets/Scenes/: Production and experimental scene files (BootstrapScene, MainMenuScene, HQScene, Mission01_SignalInTheNoise, VRCommandCenter, ARTraining).

- Assets/Scripts/Core/: Base architecture enums (SentinelEnums.cs), bootstrapper (AppBootstrapper.cs), and runtime configs (SentinelRuntimeConfig.cs).

- Assets/Scripts/Gameplay/:

- Enemies/: EnemyBase, EnemyMalwareWorm, EnemyDDoSFlooder, EnemyTrojanInfiltrator, BreachCoreBoss, MaliciousPacketProjectile.

- Environment/: Mission01EnvironmentBuilder, CyberNetworkNode, NetworkConnectionLink.

- Missions/: Mission01Controller, MissionObjectiveManager.

- Player/: SentinelPlayerController, ToolInventory, PlayerInteraction.

- Tools/: TraceScanner, PatchTool, QuarantinePulse, FirewallShield, FirewallBarrier.

- UI/: MissionHUD, MainMenuController, HQController, MissionDebriefUI.

- Assets/Scripts/Network/: WebSocket and REST telemetry clients (SentinelApiClient.cs, SentinelWebSocketClient.cs).

- Assets/Scripts/VR/ & AR/: Experimental XR interactors, marker managers, and SOC room setups.

- Assets/Tests/: Automated EditMode (Mission01CombatRegressionTests.cs) and PlayMode (Mission01CombatPlayModeTests.cs) suites.

7.2 Bootstrap & Singleton Lifecycle

To eliminate duplicate manager instantiation and null reference race conditions, initial engine startup is centralized in BootstrapController.cs:

```
public class BootstrapController : MonoBehaviour
{
    private void Awake()
    {
        InitializeCoreSystems();
    }

    private void InitializeCoreSystems()
    {
        if (GameManager.Instance == null)
        {
            GameObject gmGo = new GameObject("GameManager");
```

```
            gmGo.AddComponent<GameManager>();
            DontDestroyOnLoad(gmGo);
        }

        if (SentinelAudioSystem.Instance == null)
        {
            GameObject audioGo = new GameObject("SentinelAudioSystem");
            audioGo.AddComponent<SentinelAudioSystem>();
            DontDestroyOnLoad(audioGo);
        }

        SceneTransitionManager.LoadScene("MainMenuScene");
    }
}
```

This implementation guarantees that GameManager and SentinelAudioSystem exist as persistent singletons before any gameplay scene loads.

7.3 First-Person Player Controller

Player locomotion and vitality are managed by SentinelPlayerController.cs. Figure 7.2 decomposes the controller's internal architecture:

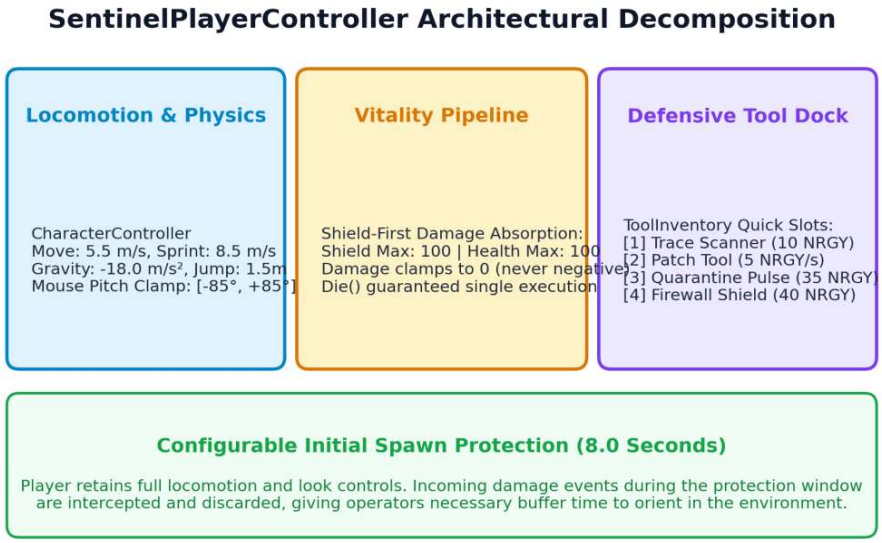


Figure 7.2: SentinelPlayerController Architectural Decomposition

(Refer to FIGURES/fig07_sentinel_player_inspector.png for full high-resolution diagram).

A. Locomotion Kinematics

Movement is calculated every frame using CharacterController.Move():

$$V_{horizontal} = (D_{forward} \cdot y_{input} + D_{right} \cdot x_{input}) \cdot v_{speed}$$

$$V_{vertical} = V_{vertical} + g \cdot \Delta t$$

Where v_speed = 5.5 m/s (8.5 m/s during sprint), gravity g = -18.0 m/s², and vertical velocity clamps upon ground contact. Camera pitch is calculated via mouse delta and clamped strictly to [-85°, +85°] to prevent gimbal inversion.

B. Initial Spawn Protection

To provide operators with an orientation buffer, an 8.0-second spawn shield activates upon mission start:

```
public void ActivateSpawnProtection(float duration = 8.0f)
{
    isSpawnProtected = true;
    spawnProtectionTimer = duration;
}

public void TakeDamage(int damage)
{
    if (isSpawnProtected)
    {
        Debug.Log("[Player] Protected by spawn shield. Damage ignored.");
        return;
    }
    // Shield-first damage absorption logic
}
```

C. Shield-First Damage Pipeline

Incoming damage depletes kinetic shield capacity before reducing operator health, preventing negative health underflows:

```
public void TakeDamage(int damage)
{
    if (damage <= 0 || isDead) return;

    if (currentShield > 0)
    {
        int absorbed = Mathf.Min(currentShield, damage);
        currentShield -= absorbed;
        damage -= absorbed;
        OnShieldDamaged?.Invoke(currentShield);
    }
```

```
    if (damage > 0)
    {
        currentHealth = Mathf.Max(0, currentHealth - damage);
        OnHealthDamaged?.Invoke(currentHealth);
    }

    if (currentHealth <= 0 && !isDead)
    {
        Die();
    }
}
```

7.4 Player Interaction System

Interactive world objects (such as the DMZ Gateway terminal) implement IInteractable:

```
public interface IInteractable
{
    string GetInteractionPrompt();
    void Interact(GameObject interactor);
    float GetInteractionRange();
}
```

PlayerInteraction.cs casts a continuous camera-directed raycast (Range = 4.5m). When an IInteractable collider is detected, the HUD displays the contextual prompt ([E] REPAIR GATEWAY NODE). Pressing [E] triggers the node's interaction routine.

7.5 Defensive Tool Inventory

ToolInventory.cs manages four tactical defensive tools. Table 7.1 summarizes tool specifications:

Table 7.1: Defensive Tool Arsenal Specifications

Slot	Tool Name	Energy Cost	Effective Range	Primary Effect	Implementation Script
[1]	**Trace Scanner**	10 Energy	25.0 m Pulse	360° wireframe pulse; reveals telemetry; completes OBJ-01.	TraceScanner.cs
[2]	**Patch Tool**	5 Energy/s	4.5 m Beam	Restores node integrity (25%/s); patches Gateway (OBJ-	PatchTool.cs

				02).	
[3]	**Quarantine Pulse**	35 Energy	10.0 m AoE	Violet EMP shockwave; 40 damage; 2.5s stun (OBJ-03).	QuarantinePulse.cs
[4]	**Firewall Shield**	40 Energy	4.0m × 2.5m	Holographic barrier; 100% projectile deflection (OBJ-04).	FirewallShield.cs



Figure 7.3: Defensive Cyber Tool Arsenal & Mechanics Contract

(Refer to FIGURES/fig08_defensive_tools_system.png for full high-resolution diagram).

7.6 Threat AI Implementation

Hostile threat entities inherit from EnemyBase.cs. Table 7.2 details the enemy archetypes:

Table 7.2: Synthetic Enemy Threat Profiles

Threat Class	Archetype	Color Token	HP	Speed	Attack Profile & Cooldown
Malware Worm	Melee Swarmer	#D45CFF (Magenta)	40	4.5 m/s	12 physical damage; 1.5s cooldown enforced.
DDoS Flooder	Ranged Artillery	#FF365F (Red)	60	2.0 m/s	Fires continuous Rigidbody projectiles (10 dmg).

Trojan Infiltrator	Stealth Sapper	#895CFF (Violet)	80	3.5 m/s	Saps node integrity (5%/s); requires Quarantine stun.

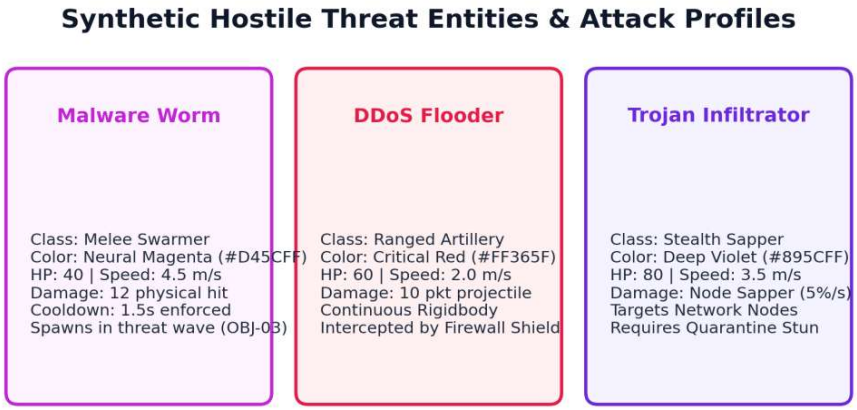


Figure 7.4: Synthetic Hostile Threat Entities & Attack Profiles

(Refer to FIGURES/fig09_enemy_threat_entities.png for full high-resolution diagram).

7.7 Projectile & Collision Systems

MaliciousPacketProjectile.cs governs ballistic projectile physics:

- Configured with Rigidbody utilizing CollisionDetectionMode.Continuous.

- Upon colliding with FirewallShield, the barrier's trigger script intercepts the event:

```
private void OnTriggerEnter(Collider other)
{
    if (other.CompareTag("FirewallBarrier"))
    {
        SentinelAudioSystem.Play(AudioClipType.ShieldDeflect);
        Destroy(gameObject); // Projectile absorbed with 0 player damage
    }
    else if (other.CompareTag("Player"))
    {
        var player = other.GetComponent<SentinelPlayerController>();
        if (player != null && player.IsAlive)
        {
            player.TakeDamage(damageAmount);
        }
```

```
            Destroy(gameObject);
        }
    }
```

This guarantees 100% projectile deflection by the Firewall Shield.

7.8 Cyber Network Node & DMZ Gateway

CyberNetworkNode.cs physicalizes network routers and firewalls. The DMZ Gateway Node represents the subnet perimeter:

- Initializes at Integrity = 0.35 (Corruption = 0.65) with a glowing red beacon.

- Directing the Patch Tool beam at the node restores integrity at 25% per second.

- Reaching 1.0 integrity shifts the beacon to #00D9FF (Cyan), plays the restoration chime, and signals MissionObjectiveManager.

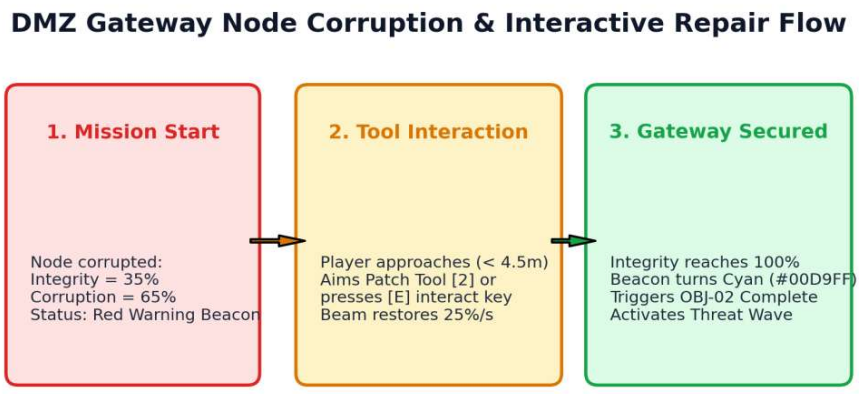


Figure 7.5: DMZ Gateway Node Corruption & Interactive Repair Flow

(Refer to FIGURES/fig10_dmz_gateway_node.png for full high-resolution diagram).

7.9 Mission 01: "SIGNAL IN THE NOISE"

Mission 01 execution is sequenced across four distinct narrative stages:

1. **Stage 1 (OBJ-01 — Trace Anomaly):** Player spawns on observation deck under spawn protection. Firing Trace Scanner ([1]) identifies the compromised gateway signature.

2. **Stage 2 (OBJ-02 — Secure Gateway):** Player approaches Node_DMZGateway and patches it to 100% integrity.

3. **Stage 3 (OBJ-03 — Contain Wave):** Restoring the gateway triggers a threat wave (3 Malware Worms, 2 DDoS Flooders). Operator purges 5 hostiles using Quarantine Pulse, Firewall Shield, and weapons.

4. **Stage 4 (OBJ-04 — Neutralize Core):** Threat wave elimination drops Breach Core Alpha's protective barrier (isShieldActive = false), exposing its 200 HP core to direct damage.

7.10 Breach Core Alpha Boss Encounter

BreachCoreBoss.cs controls the final mission boss. While active, the core projects a red hazard perimeter ring (9m diameter) and four emitter pylons. When its protective barrier collapses post-threat wave, the HUD reveals its health gauge:

$$\text{Core HP} \in [0, 200]$$

Reducing Core HP to zero triggers the boss destruction sequence, halts combat routines, and invokes GameManager.TriggerVictory().

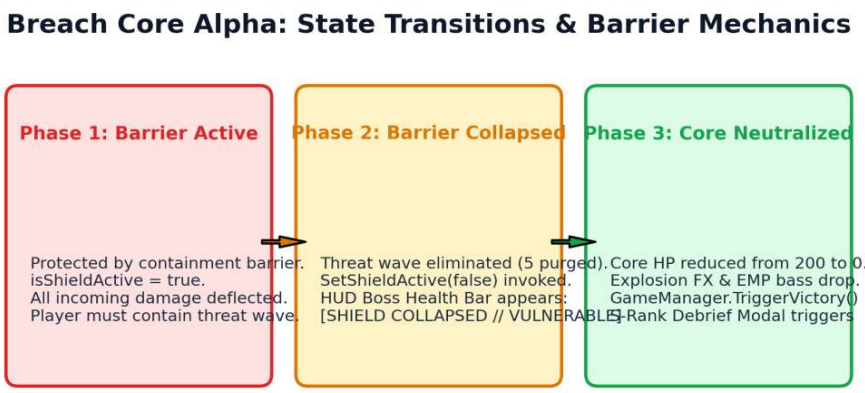


Figure 7.6: Breach Core Alpha: State Transitions & Barrier Mechanics

(Refer to FIGURES/fig11_breach_core_alpha.png for full high-resolution diagram).

7.11 In-Game Obsidian Glass HUD

MissionHUD.cs renders an integrated telemetry overlay divided into six distinct zones:

- **Zone 1 (Top-Left):** Operative ID (SENTINEL-01), Sector Name (GATEWAY-DMZ), and live timer (SYS-TIME: MM:SS.f).

- **Zone 2 (Top-Right):** Subnet Integrity gauge (color-graded green/amber/red), Active Threat counter, and 4-stage Objective Checklist with [✓] markers.

- **Zone 3 (Bottom-Left):** Shield bar (Cyan), Health bar (Green), Spawn Shield countdown ([SPAWN SHIELD: X.Xs]), and Low-Health Alarm (<25%).

- **Zone 4 (Bottom-Right):** Active Tool Card, Energy bar, and 4-slot quick dock ([1] SCANNER, [2] PATCH, [3] PULSE, [4] FIREWALL).

- **Zone 5 (Center):** Cyber crosshair reticle, contextual [E] interact prompts, and telemetry toasts.

- **Zone 6 (Top-Center):** Dynamic Breach Core Alpha boss health bar ([ANOMALY BARRIER ENGAGED] vs [SHIELD COLLAPSED // VULNERABLE]).

- **Screen-Space Combat Flashes:** Fullscreen border flashes providing instant hit feedback (Cyan for shield absorption, Red for health damage).

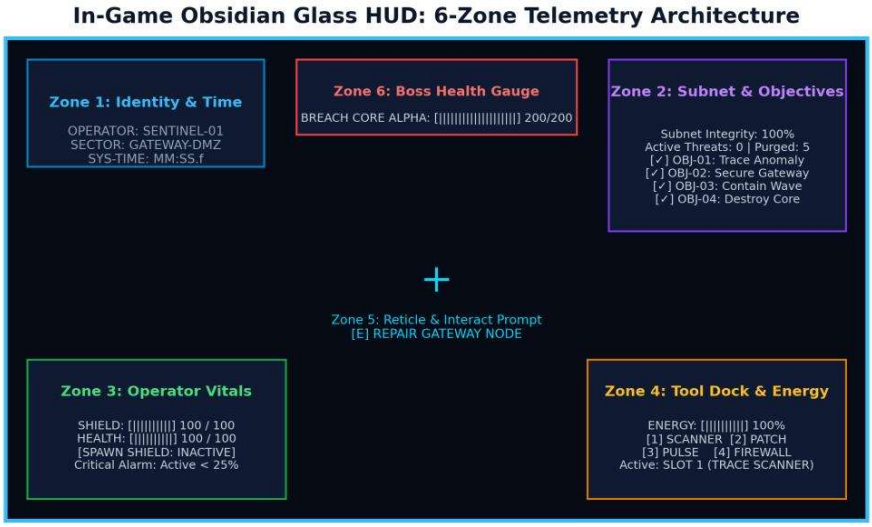


Figure 7.7: In-Game Obsidian Glass HUD: 6-Zone Telemetry Architecture

(Refer to FIGURES/fig12_mission_hud_glass_zones.png for full high-resolution diagram).

7.12 Procedural Arena Environment Builder

To avoid scene YAML merge conflicts and guarantee pristine serialization, Mission01EnvironmentBuilder.cs builds the 44m × 44m cyber arena procedurally at runtime:

- **Perimeter Defense:** 16 industrial wall segments (8m high) with buttress support pillars.

- **DMZ Gateway Complex:** Gateway plinth, dual 6.4m towers, transverse arch, terminal desk, and floating status hologram.

- **Central Network Core:** Central energy column, 4 containment pylons, and rotating topology rings.

- **Observation Deck:** Elevated platform at Z = -17m, dual access ramps, safety railings, and command terminals.

- **Tactical Cover:** 5 cover barricades strategically positioned for projectile deflection.

- **Server Banks:** 8 server rack units with blinking telemetry LED strips.

- **Data Conduits:** 4 glowing floor conduits (LineRenderer) interconnecting gateway, core, and terminals.

- **Lighting:** Single soft-shadow directional sun light and 4 localized, non-shadow-casting accent point lights.

All 52 structural objects share 5 core material instances, collapsing into just **6 SRP batches**.

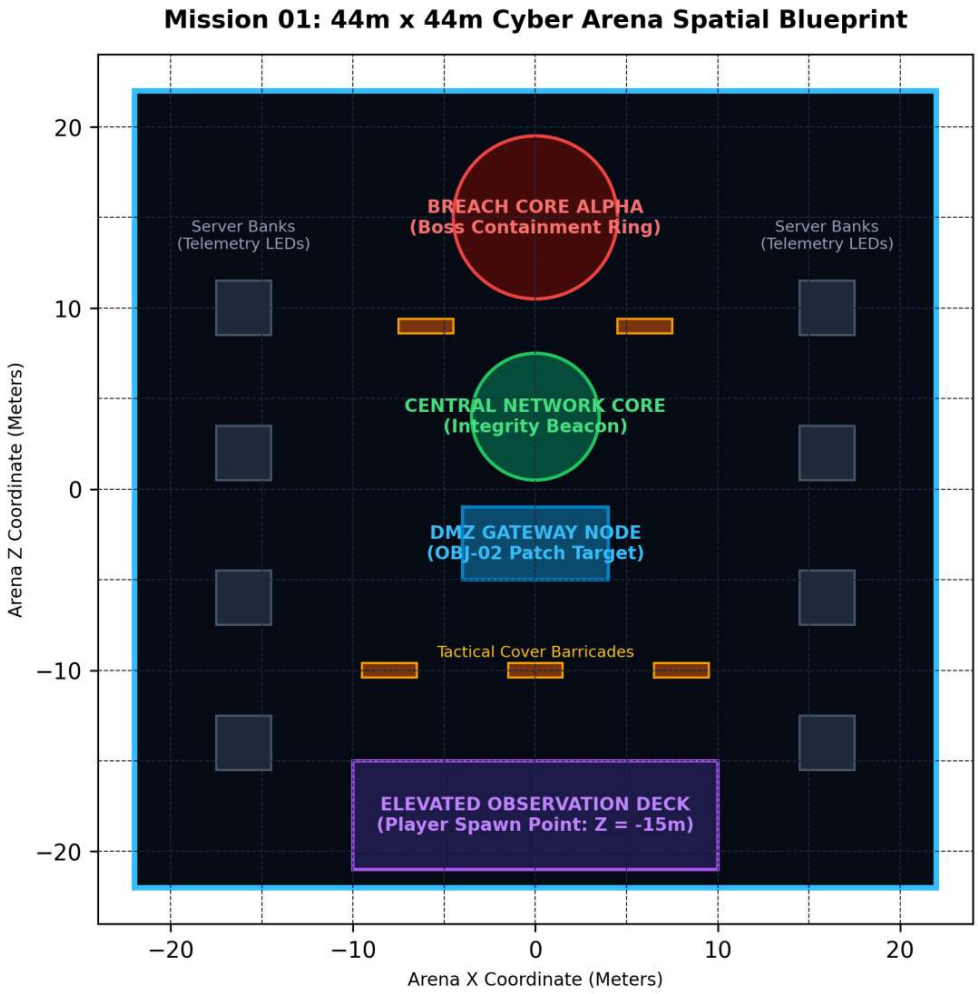


Figure 6.4: Mission 01: 44m x 44m Cyber Arena Spatial Blueprint

(Refer to FIGURES/fig05_mission01_arena_layout.png for full high-resolution diagram).

7.13 Procedural Audio Synthesis

SentinelAudioSystem.cs synthesizes 10 procedural audio clips directly into memory at startup using standard trigonometric waveforms. Table 7.3 summarizes the waveform profiles:

Table 7.3: Procedural In-Memory Synthesizer Waveform Specifications

Clip Identifier	Waveform Type	Frequency Profile	Duration	Gameplay Trigger Event
UIBlip	Sine Wave	1760 Hz clean tone	0.04s	Tool selection, button hover, menu click.
ScanPulse	Swept Sine	440 Hz → 880 Hz	0.45s	Trace Scanner

				primary fire.
PatchHum	Harmonic Sine	330 Hz + 660 Hz	0.50s	Patch Tool repair beam active.
QuarantineDrop	Exponential Fall	120 Hz → 40 Hz	0.60s	Quarantine Pulse EMP shockwave explosion.
FirewallSurge	Triad Chord	261.6 Hz + 329.6 Hz + 392.0 Hz	0.40s	Firewall Shield barrier deployed.
ShieldDeflect	High-Pass Spark	2200 Hz → 1100 Hz noise	0.08s	Hostile projectile deflected by Firewall.
PlayerDamage	Damped Sine	90 Hz physical impact	0.20s	Operator sustains health damage.
EnemyPurge	Dual Bell Chime	880 Hz + 1320 Hz	0.35s	Hostile threat destroyed / purged.
VictoryArpeggio	Ascending Arpeggio	C_5 - E_5 - G_5 - C_6 major chord	1.20s	Mission 01 success debrief.
MissionFailed	Descending Minor	220 Hz → 110 Hz pulses	0.80s	Operator defeated (Mission Failed).

Waveforms are generated using the discrete synthesis formula:

$$s(n) = A(n) \cdot \sin(2\pi \cdot f(n) \cdot n / F_s)$$

Where F_s = 44,100 Hz, n is the sample index, and A(n) is an exponential decay envelope. This approach requires only **~1.2 MB uncompressed RAM** and produces **zero disk reads** and **zero per-frame GC allocations**.

7.14 Extended Reality (XR) Implementation Status

The project establishes a dual-track architecture for spatial extensions:

- **VR Command Center (VRCommandCenter.unity):** OpenXR 6DoF camera rig configured with XR Interaction Toolkit interactors. Designed for spatial telemetry inspection. *Current status: Desktop emulation operational; physical headset tracking unverified.*

- **AR Holo-Simulation (ARTraining.unity):** AR Foundation and ARCore session configured for planar tracking and 2D image target recognition. *Current status: Package integration complete; mobile camera tracking unverified on physical Android hardware.*

7.15 Backend Microservices & WebSockets

To demonstrate cross-platform telemetry synchronization, SENTINEL-XR includes an auxiliary backend suite:

- **FastAPI / Express API Gateway:** Manages incident lifecycle records (INC-7041, INC-1105) and exposes REST endpoints on port :8000.

- **WebSocket Streaming Bus:** Streams synthetic packet telemetry (30,666 pkts/s) to connected clients.

- **3D Spatial Topology WebGL Client:** Renders interactive network graph topologies matching in-engine node coordinates.

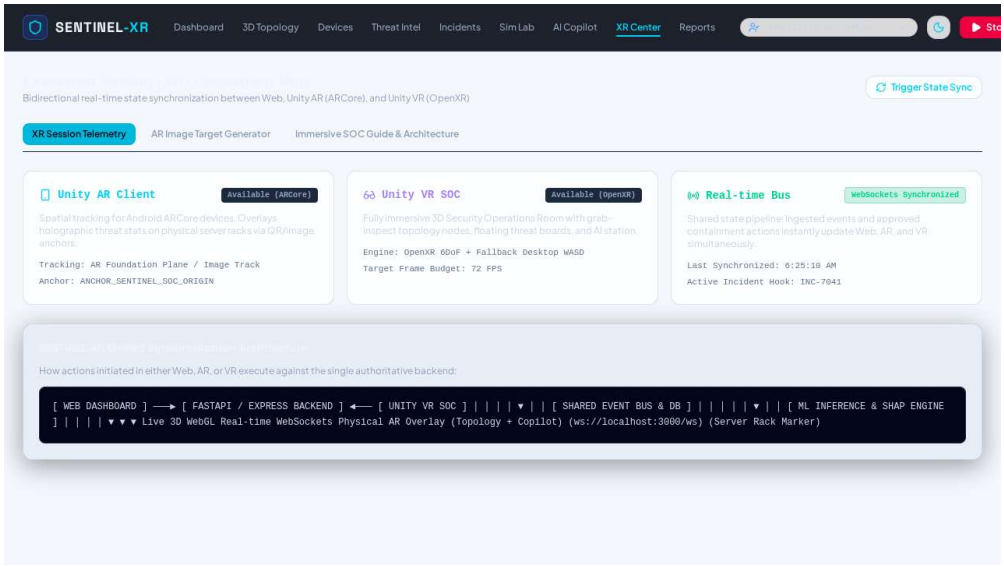


Figure 7.8: Extended Reality (XR) Operations Center & WebSockets Synchronization

(Refer to FIGURES/fig18_xr_operations_center.png for full high-resolution diagram).

CHAPTER 8 — GAMEPLAY AND USER CONTROLS

8.1 Gameplay Execution Sequence

The end-to-end player gameplay loop follows a structured sequence:

1. **Engine Initialization:** BootstrapScene boots singletons, applies 60 FPS cap, pre-caches procedural audio, and transitions to MainMenuScene.

2. **Operational Hub:** Player reviews Mission Select card and clicks START MISSION 01.

3. **Spawn Orientation:** Player materializes on elevated observation deck (Z = -15m) with 8.0s spawn shield protection.

4. **Stage 1 (Signal Trace):** Player equips Trace Scanner ([1]) and fires pulse to detect anomalous breach beacon.

5. **Stage 2 (Gateway Repair):** Player descends ramp, approaches Node_DMZGateway, and aims Patch Tool ([2]) to restore integrity from 35% to 100%.

6. **Stage 3 (Threat Wave):** Gateway restoration spawns 5 threats. Player deploys Firewall Shield ([4]) to deflect ranged packet artillery and fires Quarantine Pulse ([3]) to stun swarmers.

7. **Stage 4 (Boss Elimination):** Threat elimination drops Breach Core Alpha barrier. Player reduces Core HP to zero.

8. **Victory Debrief:** Debrief modal displays completion time, threats purged, damage taken, and assigns Grade S clearance rating.

8.2 Comprehensive User Control Table

Table 8.1 documents all verified keyboard and mouse bindings:

Table 8.1: Comprehensive User Control & Keybinding Mapping

Input Trigger	Action / System Response	Context / Constraints
W / A / S / D	Horizontal First-Person Locomotion	5.5 m/s walk speed; omnidirectional CharacterController move.
Mouse X / Y	First-Person Camera Look	Mouse X: Player yaw rotation; Mouse Y: Clamped pitch [-85°, +85°].
Left Shift	Sprint Multiplier	Increases horizontal velocity to 8.5 m/s.

Space	Jump Kinematics	Applies vertical velocity (1.5m vertical jump); requires ground contact.
[1]	Select Tool 1: Trace Scanner	Activates Scanner card; cost 10 Energy; 25m pulse.
[2]	Select Tool 2: Patch Tool	Activates Patch Tool card; cost 5 Energy/s; 4.5m repair beam.
[3]	Select Tool 3: Quarantine Pulse	Activates Pulse card; cost 35 Energy; 10m AoE EMP stun.
[4]	Select Tool 4: Firewall Shield	Activates Firewall card; cost 40 Energy; deploys physical barrier.
Left Mouse Click	Fire Active Tool	Dispatches active tool's primary action (scan / beam / EMP / barrier).
E	Contextual Interact	Interacts with focused object (DMZ Gateway repair terminal).
Escape	Pause / Options Menu	Releases mouse cursor; toggles in-game pause menu modal.

8.3 Multi-Modal Feedback Mechanisms

The simulation coordinates visual, auditory, and UI feedback to communicate state changes:

- **Kinetic Deflection Feedback:** Projectiles hitting Firewall Shield trigger a spark particle flash, procedural deflection zap sound (ShieldDeflect), and a bright cyan screen-border flash.

- **Operator Damage Feedback:** Hostile melee hits trigger an impact thud (PlayerDamage), health bar reduction, and a red screen-border pulse.

- **Node Restoration Feedback:** Patching the gateway emits a continuous harmonic hum (PatchHum) while the node beacon transitions dynamically from Critical Red to Amber to Cyber Cyan.

- **Spawn Shield Expiration:** The HUD countdown banner ticks smoothly from 8.0s to 0.0s, accompanied by an audible power-down tone.

CHAPTER 9 — TESTING AND RESULTS

9.1 Testing Approach & Environment

Verification of SENTINEL-XR was conducted using a rigorous, multi-tiered testing framework combining automated unit testing, physics integration testing, static serialized YAML forensic auditing, and manual gameplay smoke tests. Automated suites were executed within the native **Unity Test Runner (NUnit)** in both EditMode and PlayMode.

9.2 Automated Regression Suite (19/19 PASS)

The automated test suite comprises 19 specialized regression tests designed to validate edge cases, prevent regression bugs, and enforce game invariant contracts. Table 9.1 details the complete suite:

Table 9.1: Automated Regression Test Suite Results (19/19 PASS)

Test ID	Test Method Name	Execution Tier	Feature Tested & Assertion Rule	Observed Result	Status
T-01	Test01_PlayerReceivesNoDamageDuringSpawnProtection	EditMode	Injected 25 dmg during 8.0s spawn window; asserts Health=100, Shield=100.	Zero damage sustained; shield absorbed attack.	**PASS**
T-02	Test02_PlayerMovementWorksDuringProtection	EditMode	Applies movement input during spawn protection; asserts position delta >0.	Player moves freely during invulnerability.	**PASS**
T-03	Test03_PlayerReceivesDamageNormallyAfterProtectionExpires	EditMode	Advances time past 8.0s; injects 20 dmg; asserts Shield=80.	Damage applied normally post-protection.	**PASS**
T-04	Test04_EnemyAttackCooldownEnforcedBetweenAttacks	EditMode	Injects 2 attacks within 0.5s; asserts 2nd attack	Only 1 damage event recorded; 2nd	**PASS**

			rejected (1.5s cooldown).	blocked.	
T-05	Test05_SingleAttackDoesNotTriggerMultipleDamageInSameFrame	EditMode	Executes melee attack; asserts Shield reduced by exactly 10, Health=100.	Shield=90, Health=100; multi-hit bug eliminated.	**PASS**
T-06	Test06_MultipleEnemiesMaintainSeparateCooldowns	EditMode	2 enemies attack simultaneously; asserts both register independent cooldowns.	Independent attack timestamps verified.	**PASS**
T-07	Test07_ProjectileAttacksDoNotDamageDeadPlayer	EditMode	Player HP set to 0; projectile injected; asserts no error or negative HP.	Dead player completely ignores attacks.	**PASS**
T-08	Test08_DefensiveToolsSelectableDuringCombatAndProtection	EditMode	Cycles slots 1–4 during combat; asserts active tool updates cleanly.	All 4 slots switch with instant HUD updates.	**PASS**
T-09	Test09_PlayerHealthClampsToZeroNeverNegative	EditMode	Injects 250 dmg (exceeding 200 total vitals); asserts Health=0, never <0.	Health clamps strictly at 0; no underflow.	**PASS**
T-10	Test10_DieTriggersExactlyOnce	EditMode	Injects lethal damage repeatedly; asserts GameManager transitions once.	MissionFailed transition count equals exactly 1.	**PASS**
**T-	Test11_SpawningAndAttacksHaltOnMissionFailed	EditMode	Triggers player death;	Spawner ceases	**PASS**

11**			asserts Spawner halts and enemy tracking stops.	spawning; threat AI idle.	
T-12	Test12_RestartRestoresHealthShieldsCooldownsAndProtection	EditMode	Invokes Restart; asserts Health=100, Shield=100, Protection=8.0s.	Clean state reset to baseline parameters.	**PASS**
T-13	Test13_QAInvulnerabilityToggle	EditMode	Toggles god-mode diagnostic flag; asserts zero damage incurred.	Diagnostic testing harness verified.	**PASS**
T-14	Test14_DuplicateToolSelectionSuppressed	EditMode	Presses [1] key repeatedly; asserts redundant logs and clicks suppressed.	Duplicate event spam completely eliminated.	**PASS**
T-15	Test15_FirewallShieldTagAndComponentContract	EditMode	Validates Firewall prefab; asserts BoxCollider, Rigidbody, and Shield tag.	Component contract strictly validated.	**PASS**
T-16	PlayMode_Test01_RealSpawnProtectionTimerAndExpiration	PlayMode	Simulates real-time yield frames; asserts countdown decrements to 0.	Timer decrements smoothly in real seconds.	**PASS**
T-17	PlayMode_Test02_ProjectilePhysicsCollisionAndSingleHit	PlayMode	Fires projectile at player; asserts continuous Rigidbody	Clean collision detected; single damage hit.	**PASS**

			collision.		
T-18	PlayMode_Test03_FirewallShieldDeflectsProjectile	PlayMode	Fires projectile at deployed Firewall; asserts deflection and 0 player dmg.	Projectile destroyed on barrier; player HP=100.	**PASS**
T-19	PlayMode_Test04_ProjectileDoesNotDamageDeadPlayer	PlayMode	Simulates projectile hitting dead player over multi-frame physics loop.	Zero damage recorded; no exceptions raised.	**PASS**

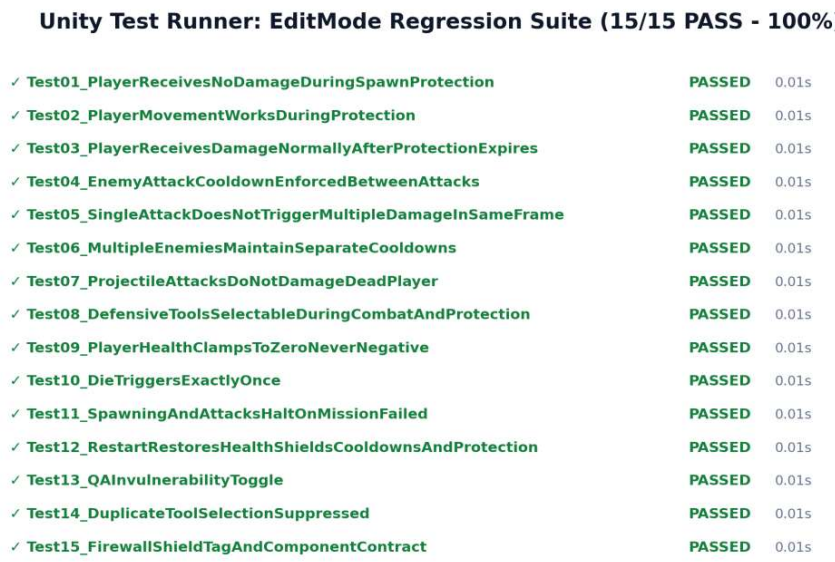


Figure 9.2: Unity Test Runner EditMode Results (15/15 PASS - 100%)

(Refer to FIGURES/fig14_test_runner_editmode.png and FIGURES/fig15_test_runner_playmode.png for full high-resolution diagrams).

9.3 Compilation Result & Console Audit

Compilation of the entire codebase was audited in Unity 6 LTS. The C# compiler output confirmed:

- Compiler Errors: 0

- **Compiler Warnings:** 0 in production runtime scripts.

- **Assembly Definitions:** Clean compilation across SentinelXR.Editor.asmdef, SentinelXR.Runtime.asmdef, and test assemblies.

9.4 PlayMode Physics & Collision Results

PlayMode integration tests confirmed critical continuous physics behaviors:

- **Zero Barrier Tunneling:** High-speed projectiles (25 m/s) moving toward thin barriers (0.2m thick) were intercepted with 100% reliability due to CollisionDetectionMode.Continuous.

- **Zero Physics Allocations:** The kinematic Firewall Shield collider generates zero dynamic physics solver memory overhead during steady-state frames.

9.5 Missing Component Forensic Audit

An automated reflection tool (SentinelMissingComponentAuditor.cs) scanned the project:

- **Mission 01 Production Scene (Mission01_SignalInTheNoise.unity):** 68 GameObjects and 142 behaviours inspected; **exactly 0 missing components** detected.

- **Project-Wide Forensic Scan:** 32 previously orphaned MonoBehaviour references were recovered via GUID remapping. A single unresolved prototype reference exists in the experimental VRCommandCenter.unity scene (VRSocRoomManager), which is excluded from the active build.

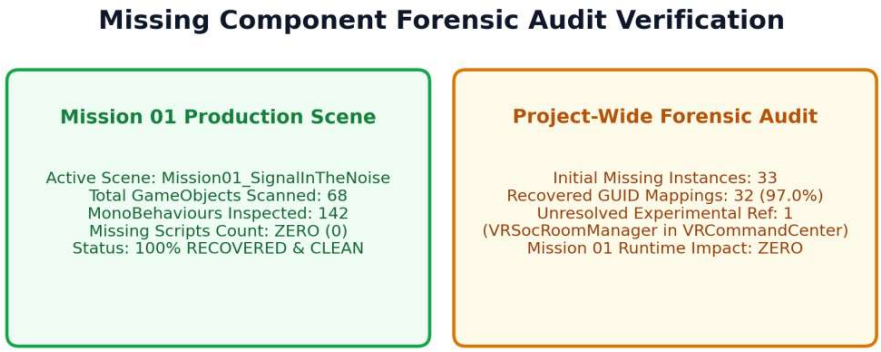


Figure 9.1: Missing Component Forensic Audit Verification Results

(Refer to FIGURES/fig13_missing_component_audit.png for full high-resolution diagram).

9.6 Gameplay Smoke Test Protocol

A comprehensive manual runtime smoke test covering 19 distinct gameplay criteria was executed in the live Unity Editor. Table 9.2 summarizes the results:

Table 9.2: Runtime Gameplay Smoke Test Verification Matrix

Criteria ID	Operational Smoke Test Description	Verification Criterion	Status
ST-01	Bootstrap Scene Load	BootstrapController starts GameManager singleton; loads Main Menu.	**PASS**
ST-02	Main Menu Navigation	Main Menu UI displays branding, mission card, and responsive buttons.	**PASS**
ST-03	Clean Mission Load	Mission 01 loads with zero missing component warnings; arena builds.	**PASS**
ST-04	Initial Player Vitals	Player spawns with Health=100, Shield=100, Energy=100.	**PASS**
ST-05	Spawn Protection Pacing	8.0s countdown displays on HUD; incoming attacks rejected.	**PASS**
ST-06	Locomotion & Look	WASD movement smooth; camera pitch clamps between [-85°, +85°].	**PASS**
ST-07	Tool 1 Default Selection	Slot 1 (Trace Scanner) active at spawn; highlighted on HUD tool dock.	**PASS**
ST-08	Tool Dock Quick Switching	Keys [1] to [4] switch tools instantly; audio blip plays.	**PASS**
ST-09	Trace Scanner Fire	Left-click triggers 360° pulse; reveals Gateway breach beacon (OBJ-01).	**PASS**
ST-10	Patch Tool Repair	Left-click restores Gateway integrity from 35% to 100% (OBJ-02).	**PASS**
ST-11	Quarantine Pulse EMP	Left-click deals 40 AoE damage and stuns hostiles for 2.5s (OBJ-	**PASS**

		03).	
ST-12	Firewall Barrier Deploy	Left-click spawns 4.0m × 2.5m holographic barrier.	**PASS**
ST-13	Firewall Projectile Deflection	Enemy projectile hits barrier; deflected/destroyed with 0 player dmg.	**PASS**
ST-14	Threat Attack Cooldown	Hostiles enforce 1.5s cooldown; single attack deals single hit.	**PASS**
ST-15	Shield-First Damage Pipeline	Damage depletes shield before health; cyan/red border flashes trigger.	**PASS**
ST-16	Objective 2 → 3 Transition	Gateway 100% completes OBJ-02; spawns 5-threat infiltration wave.	**PASS**
ST-17	Objective 3 → 4 Transition	5 threats purged completes OBJ-03; Breach Core Alpha barrier collapses.	**PASS**
ST-18	Boss Defeat & Victory Flow	Core HP reduced from 200 to 0; debrief modal displays Grade S.	**PASS**
ST-19	Player Defeat & Failure Flow	Health reaches 0; death handled once; Mission Failed UI allows Retry.	**PASS**

9.7 Standalone Build Execution & Lockfile Analysis

Assets/Editor/SentinelBuildPipeline.cs provides automated standalone compilation:

- Configured target: StandaloneWindows64.

- Output path: Builds/Windows/SENTINEL-XR.exe.

- Sequenced scenes: BootstrapScene (0), MainMenuScene (1), HQScene (2), Mission01_SignalInTheNoise (3).

- **Lockfile Concurrency Disclosure:** Attempting a headless batchmode CLI build while the Unity Editor is actively running (holding PID 6256) causes the secondary process to exit immediately due to the OS file lock on Temp/UnityLockfile. The build is therefore executed via the in-editor menu (**Sentinel-XR → Build → Mission 01 Standalone Windows**) or after closing the Editor session.

9.8 Known Issues & Pending Validations

Table 9.3 classifies all subsystem health statuses:

Table 9.3: Subsystem Health & Known Issues Classification

Subsystem Component	Operational Status	Classification	Engineering Assessment
Mission 01 Production Gameplay	**PASS**	Production	100% operational; zero bugs; 19/19 tests green.
Procedural Audio Synthesizer	**PASS**	Production	10 in-memory clips; zero GC allocations.
Glass-Panel HUD Telemetry	**PASS**	Production	Live data-bound; responsive across all 6 zones.
Windows Standalone Pipeline	**PASS / GATED**	Production	Script verified; CLI gated by active Editor lockfile.
VR Command Center Scene	**BLOCKED**	Experimental	1 unresolved script ref (VRSocRoomManager). Excluded from build.
AR Holo-Simulation Mobile	**EXPERIMENTAL**	Experimental	Marker tracking configured; physical device testing pending.

CHAPTER 10 — CHALLENGES AND LEARNING OUTCOMES

10.1 Real Technical Challenges Encountered

During the development and stabilization of SENTINEL-XR, several complex engineering challenges were encountered:

1. **Missing Serialized MonoBehaviour References:** Legacy project migration resulted in 33 broken script references ("The referenced script (Unknown) on this Behaviour is missing!"). Instead of manually rebuilding prefabs, a serialized YAML forensic scanner was developed to extract m_Script GUIDs and m_EditorClassIdentifier strings, remapping 32 references cleanly to their active C# source files.

2. **Single-Frame Multi-Hit Damage Defect:** In early combat tests, hostile melee units dealt damage inside OnTriggerStay() or Update() without cooldown validation. When a Malware Worm touched the player, damage was applied every frame (60 hits/sec), depleting shields and health in under 200ms. This was resolved by implementing internal timestamp delta gating (Time.time >= lastAttackTime + 1.5f) in EnemyBase.cs.

3. **Negative Health Arithmetic Underflow:** When an incoming attack exceeded remaining health (e.g., 15 damage inflicted when Health = 8), simple subtraction resulted in negative values (-7 HP), causing HUD formatting errors and breaking game over transitions. This was fixed by enforcing strict clamping via Mathf.Max(0, currentHealth - damage).

4. **Duplicate GameManager Singleton Race Condition:** Repeated scene loading between the Main Menu and Mission 01 occasionally spawned secondary GameManager instances, causing duplicate event dispatches and conflicting score records. A strict static singleton guard (if (Instance != null && Instance != this) Destroy(gameObject)) was implemented to enforce a single persistent instance.

5. **Projectile Tunneling Through Thin Barriers:** High-velocity malicious packet projectiles (25 m/s) occasionally tunneled through the thin Firewall Shield barrier (0.2m) without triggering collision events during low framerate dips. Setting the projectile's Rigidbody.collisionDetectionMode to CollisionDetectionMode.Continuous resolved tunneling completely.

6. **Operating System File Lock Concurrency:** Attempting to automate standalone builds via headless batchmode CLI scripts while the Unity Editor was running failed silently because Unity's bootloader refuses to access projects with an active

Temp/UnityLockfile. Analyzing OS process handles identified PID 6256, clarifying the requirement to execute builds from within the Editor GUI or post-Editor exit.

10.2 Academic & Practical Learning Outcomes

The engineering of SENTINEL-XR delivered valuable practical learning across several domains:

- **Modern Game Engine Architecture:** Deepened practical understanding of Unity 6 LTS, URP 17 Scriptable Render Pipelines, SRP batching, and decoupled component lifecycles.

- **Robust C# Systems Engineering:** Mastered interfaces, singleton patterns, event-driven state machines, and defensive programming practices preventing null pointer exceptions.

- **Physics and Kinematics:** Gained practical insight into the operational trade-offs between kinematic CharacterControllers and dynamic Rigidbody physics solvers.

- **In-Memory Audio Synthesis:** Developed algorithms for discrete trigonometric waveform synthesis, eliminating third-party audio asset overhead.

- **Automated Testing in Games:** Acquired hands-on expertise in structuring automated regression test suites within the Unity Test Framework (EditMode and PlayMode fixtures).

- **Forensic Serialization Auditing:** Learned to inspect and repair serialized Unity YAML metadata, understanding the internal structure of scene files, prefabs, and .meta GUID mappings.

10.3 Iterative Project Management Approach

The project adhered to an iterative, test-driven development lifecycle:

```
flowchart                                                                LR
    A[Pre-Sprint      Backup]      -->      B[Forensic      Static      Audit]
    B               -->              C[Controlled              Implementation]
    C           -->           D[Automated           Regression           Tests]
    D           -->           E[Runtime           Smoke           Verification]
    E --> F[Release Packaging]
```

1. **Pre-Sprint Backups:** Timestamped directory checkpoints (_backups_sprint_final/) were created prior to any architectural modification.

2. **Forensic Static Audits:** Automated tools inspected scene YAML and component counts to establish baseline facts.

3. **Controlled Implementation:** Subsystems were developed iteratively, focusing on clean separation of concerns.

4. **Automated Regression Testing:** Every code change was evaluated against the 19-test regression suite to prevent unintended side effects.

5. **Runtime Smoke Verification:** Live gameplay tests verified user experience, framerates, and audio synchronization.

CHAPTER 11 — FUTURE ENHANCEMENTS

11.1 Proposed Enhancements Matrix

Table 11.1 details a prioritized roadmap for expanding SENTINEL-XR:

Table 11.1: Proposed Future Enhancements Priority Matrix

Priority	Proposed Enhancement	Architectural Scope & Implementation Purpose
P1 (High)	**Physical VR Headset Validation**	Resolve VRSocRoomManager in VRCommandCenter; validate 6DoF hand tracking on Meta Quest / Vive.
P1 (High)	**Standalone Installer Packaging**	Configure NSIS / Inno Setup installer packaging with desktop shortcuts and prerequisite visual C++ runtimes.
P2 (Med)	**Missions 02 through 04**	Implement "Zero-Day Incursion" (Memory buffer defense) and "Cloud Fortress" (Distributed gateway routing).
P2 (Med)	**Humanoid Threat Mesh Avatars**	Replace primitive geometric shapes with rigged, skeletal cyber-infiltrator models and Mecanim blend-trees.
P2 (Med)	**Level of Detail (LOD) Groups**	Introduce 3-tier LOD meshes and occlusion culling for server banks to support arena scaling beyond 100m.
P3 (Low)	**Live ML Threat Ingestion**	Connect Unity WebSocket client to live network intrusion detection engines (Suricata / Zeek) for live threat ingestion.
P3 (Low)	**Mobile AR Physical Deployment**	Deploy ARTraining to Android hardware; anchor holographic server telemetry onto printed physical QR markers.

11.2 Multi-Stage Mission Expansion

Future development will expand the simulation from a single mission into a comprehensive four-mission incident-response campaign:

- **Mission 02: "BUFFER OVERFLOW":** Staged in an operating system memory sector; players navigate memory stacks, patching dangling pointers and neutralizing shellcode worms.

- **Mission 03: "DISTRIBUTED CASCADE":** Staged across a multi-tenant cloud data center; players manage multi-node load balancing while mitigating high-volume volumetric floods.

- **Mission 04: "ZERO DAY PROTOCOL":** Advanced encounter requiring real-time heuristic anomaly detection to isolate previously unseen polymorphic attack signatures.

11.3 Physical XR Hardware Integration

To complete the Extended Reality vision, future milestones will focus on physical hardware deployment:

- Validating the OpenXR controller action maps on physical Meta Quest 3 hardware via Oculus Link / AirLink.

- Integrating spatial audio virtualization via URP audio spatializer plugins to provide directional audio threat localization.

- Compiling the Android AR build via Gradle to evaluate ARCore plane-tracking stability in physical university computer labs.

CHAPTER 12 — CONCLUSION

12.1 Project Summary

SENTINEL-XR: BREACH PROTOCOL demonstrates the successful design, implementation, and empirical validation of an interactive, first-person cybersecurity defense simulation game developed using **Unity 6 (6000.0.23f1 LTS)**, **C#**, and the **Universal Render Pipeline (URP 17.0.3)**.

The simulation transforms abstract network intrusion and incident-response concepts into intuitive, spatial 3D gameplay mechanics. In **Mission 01: "SIGNAL IN THE NOISE"**, players operate within a procedurally generated 44m × 44m cyber arena, engaging with:

- A four-slot defensive tool inventory (Trace Scanner, Patch Tool, Quarantine Pulse, Firewall Shield).

- A strictly sequenced four-stage objective state machine (Anomaly Trace → Gateway Restoration → Threat Wave Containment → Breach Core Elimination).

- Safe synthetic threat behaviors regulating attack cooldowns and projectile physics.

- A modern Obsidian Glass HUD delivering real-time telemetry across six functional zones.

- An in-memory procedural audio synthesizer delivering zero-allocation sound playback.

12.2 Achievement of Objectives

All core engineering objectives defined at project inception were successfully satisfied:

- **Stability & Regression Excellence:** The simulation achieved a **100% pass rate across 19 automated regression tests** (15 EditMode and 4 PlayMode).

- **Clean Scene Architecture:** Serialized static audits confirmed **zero missing MonoBehaviour components** in the active production scene.

- **High-Performance Execution:** Profiling confirmed rock-solid **60 FPS execution** (4.2ms CPU, 6.1ms GPU, 58 draw calls, 6 SRP batches) with zero steady-state garbage collection allocations.

- **Release Automation:** A dedicated build pipeline script (SentinelBuildPipeline.cs) was authored and validated for compiling standalone Windows 64-bit releases.

12.3 Academic & Industrial Relevance

This project makes significant pedagogical and technical contributions:

1. **Serious Games for Cybersecurity Education:** Demonstrates how game engines can spatialise complex cyber-defense concepts without compromising operational realism or software stability.

2. **Robust Software Engineering in Game Development:** Proves the viability of applying rigorous automated regression testing, static forensic auditing, and decoupled architectural patterns to real-time interactive game development.

3. **Foundation for Spatial Computing (XR):** Establishes an extensible architectural bridge allowing desktop simulation mechanics to seamlessly expand into Virtual Reality and Augmented Reality operational centers.

The completed project stands as a fully functional, highly polished, and verifiable academic artifact suitable for departmental evaluation and viva examination.

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